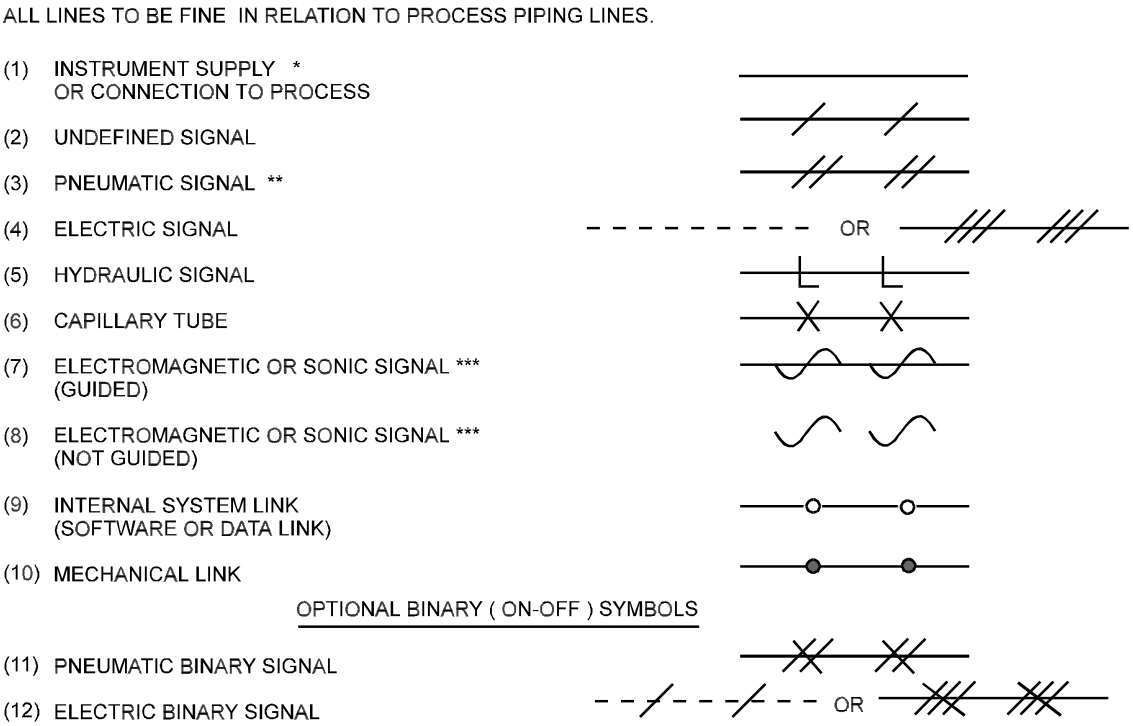
6.2 Instrument line symbols

ALL LINES TO BE FINE IN RELATION TO PROCESS PIPING LINES.



NOTE: 'OR' means user's choice. Consistency is recommended.

* The following abbreviations are suggested to denote the types of power supply. These designations may also be applied to purge fluid supplies.

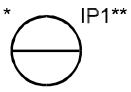
AS - Air Supply
IA - Instrument Air
PA - Plant Air
} Options
ES - Electric Supply
GS - Gas Supply
HS - Hydraulic Supply
NS - Nitrogen Supply
SS - Steam Supply
WS - Water Supply

The supply level may be added to the instrument supply line, e.g., AS-100, a 100 psig air supply; ES-24-DC, a 24-volt direct current power supply.

** The pneumatic signal symbol applies to a signal using any gas as the signal medium. If a gas other than air is used, the gas may be identified by a note on the signal symbol or otherwise.

*** Electromagnetic phenomena include heat, radio waves, nuclear radiation and light.

6.3 General instrument or function symbols

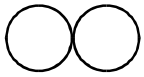
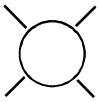
	PRIMARY LOCATION ***NORMALLY ACCESSIBLE TO OPERATOR	FIELD MOUNTED	AUXILIARY LOCATION *** NORMALLY ACCESSIBLE TO OPERATOR
DISCRETE INSTRUMENTS	1 	2 	3 
SHARED DISPLAY, SHARED CONTROL	4 	5 	6 
COMPUTER FUNCTION	7 	8 	9 
PROGRAMMABLE LOGIC CONTROL	10 	11 	12 

* Symbol size may vary according to the user's needs and the type of document. A suggested square and circle size for large diagrams is shown above. Consistency is recommended.

** Abbreviations of the user's choice such as IP1 (Instrument Panel #1), IC2 (Instrument Console #2), CC3 (Computer Console #3), etc., may be used when it is necessary to specify instrument or function location.

*** Normally inaccessible or behind-the-panel devices or functions may be depicted by using the symbol but with dashed horizontal bars, i.e. 

6.3 General instrument or function symbols (contd.)

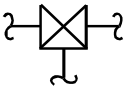
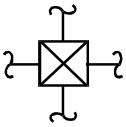
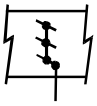
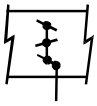
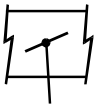
13	14  INSTRUMENT WITH LONG TAG NUMBER	15  INSTRUMENTS SHARING COMMON HOUSING *
16  PILOT LIGHT	17  PANEL MOUNTED PATCHBOARD POINT 12	18 **  PURGE OR FLUSHING DEVICE
19 **  REST FOR LATCH-TYPE ACTUATOR	20  DIAPHRAGM SEAL	21 ** ***  UNDEFINED INTERLOCK LOGIC

* It is not mandatory to show a common housing.

** These diamonds are approximately half the size of the larger ones.

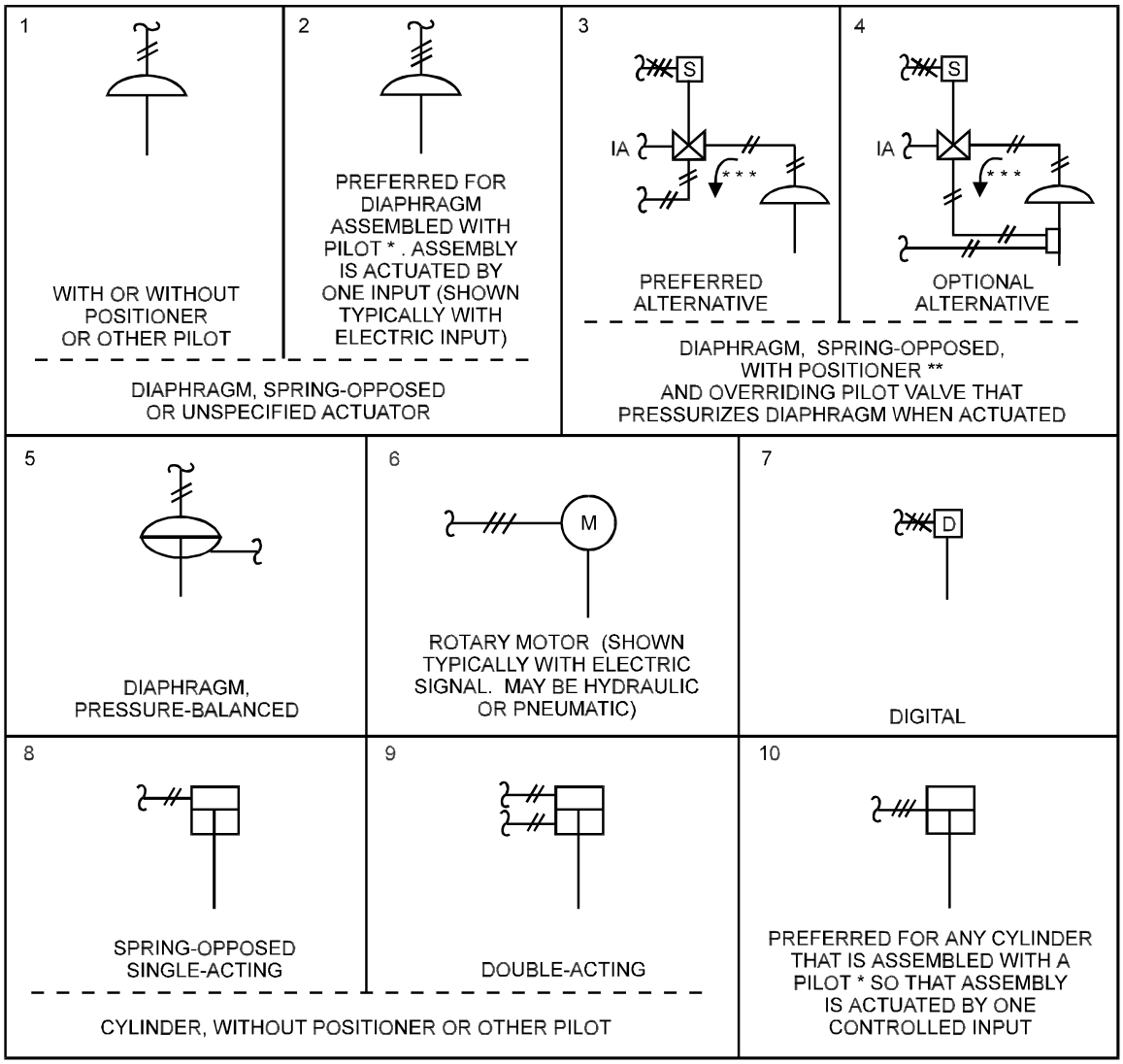
*** For specific logic symbols, see ANI/ISA Standard S5.2.

6.4 Control valve body symbols, damper symbols

1  GENERAL SYMBOL	2  ANGLE	3  BUTTERFLY	4  ROTARY VALVE
5  THREE-WAY	6  FOUR-WAY	7  GLOBE	8
9  DIAPHRAGM	10 	11 	12 
	DAMPER OR LOUVER		

Further information may be added adjacent to the body symbol either by note or code number.

6.5 Actuator symbols

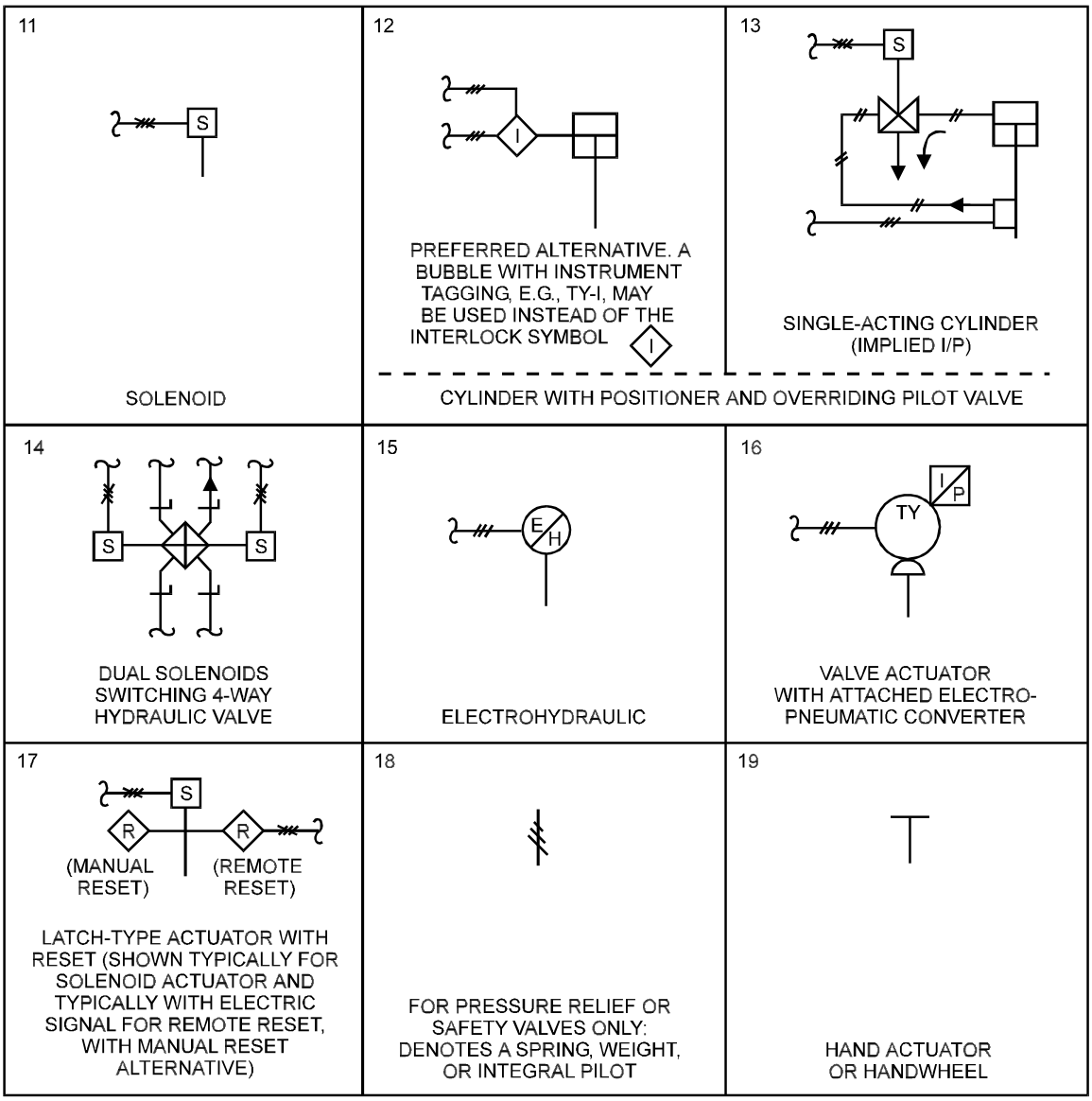


* Pilot may be positioner, solenoid valve, signal converter, etc.

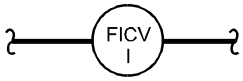
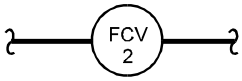
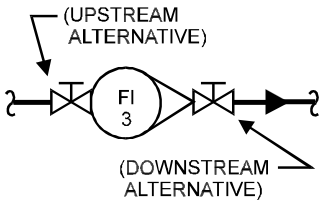
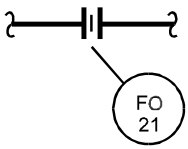
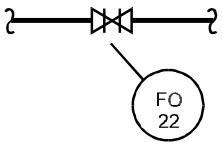
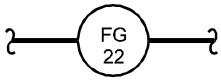
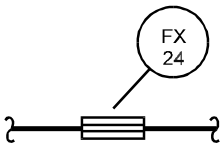
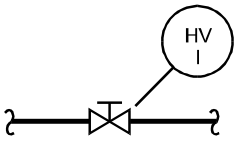
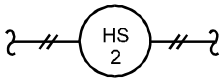
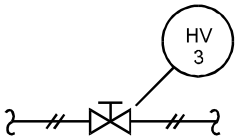
* * The positioner need not be shown unless an intermediate device is on its output. The positioner tagging, ZC, need not be used even if the positioner is shown. The positioner symbol, a box drawn on the actuator shaft, is the same for all types of actuators. When the symbol is used, the type of instrument signal, i.e., pneumatic, electric, etc., is drawn as appropriate, if the positioner symbol is used and there is no intermediate device on its output, then the positioner output signal need not be shown.

* * * The arrow represents the path from a common to a fail open port. It does not correspond necessarily to the direction of fluid flow.

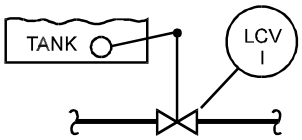
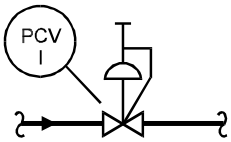
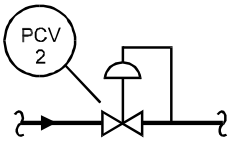
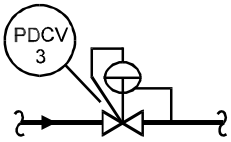
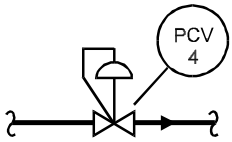
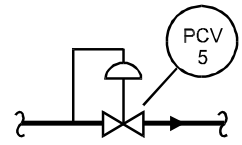
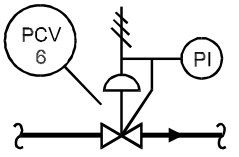
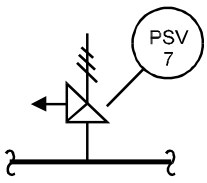
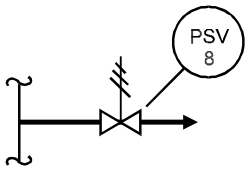
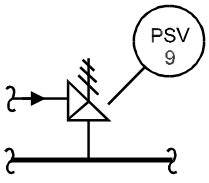
6.5 Actuator symbols (contd.)



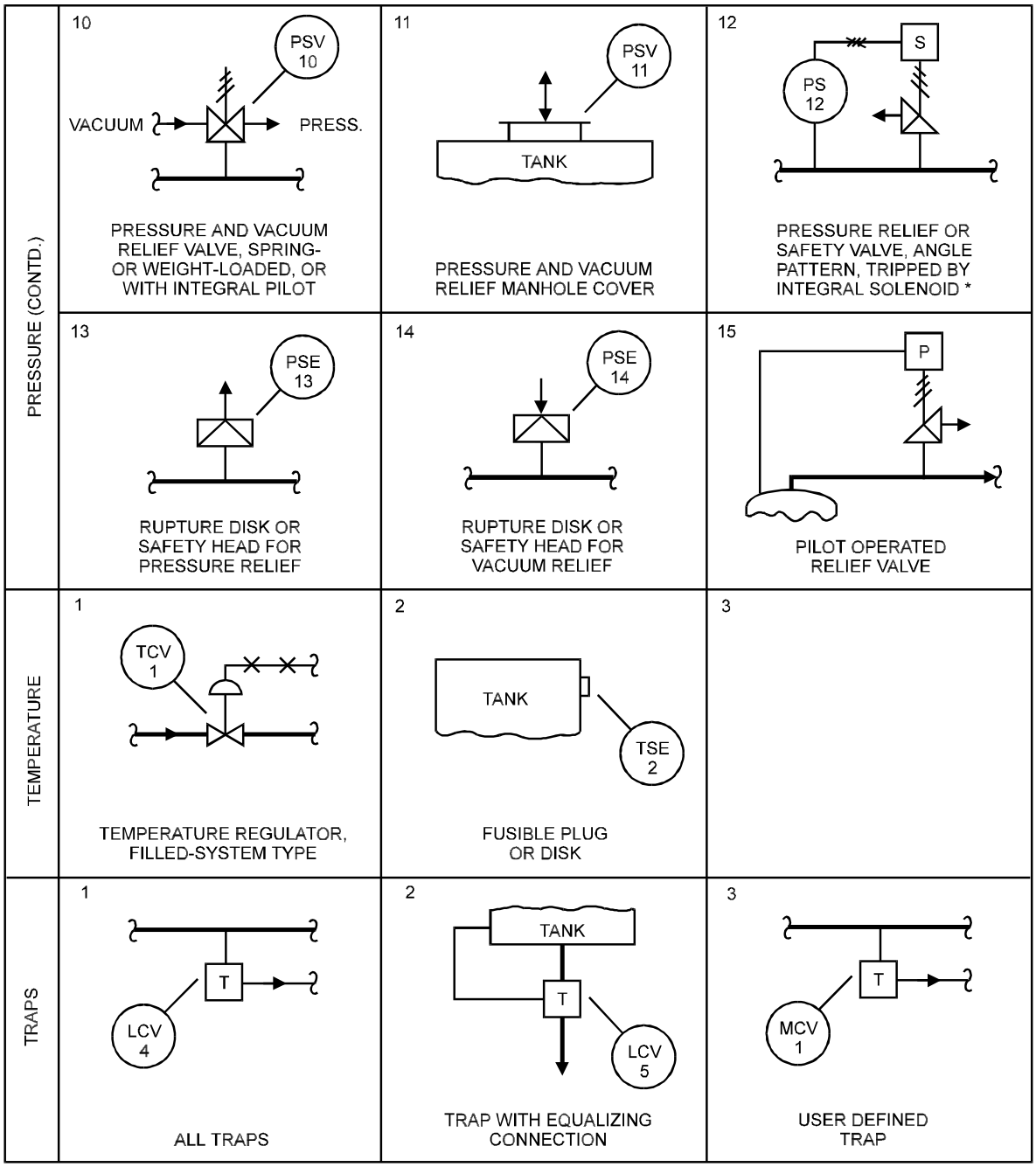
6.6 Symbols for self-actuated regulators, valves, and other devices

FLOW	1  AUTOMATIC REGULATOR WITH INTEGRAL FLOW INDICATION	2  AUTOMATIC REGULATOR WITHOUT INDICATION	3  INDICATING VARIABLE AREA METER WITH INTEGRAL MANUAL THROTTLE VALVE
	4  RESTRICTION ORIFICE (ORIFICE PLATE, CAPILLARY TUBE OR MULTI-STAGE TYPE, ETC.) IN PROCESS LINE	5  RESTRICTION ORIFICE DRILLED IN VALVE (INSTRUMENT TAG NUMBER MAY BE OMITTTED IF VALVE IS OTHERWISE IDENTIFIED)	6  FLOW SIGHT GLASS, PLAIN OR WITH PADDLE WHEEL, FLAPPER, ETC.
	7  FLOW STRAIGHTENING VANE (USE OF TAG NUMBER IS OPTIONAL. THE LOOP NUMBER MAY BE THE SAME AS THAT OF THE ASSOCIATED PRIMARY ELEMENT)	8	9
HAND	1  HAND CONTROL VALVE IN PROCESS LINE	2  HAND-ACTUATED ON-OFF SWITCHING VALVE IN PNEUMATIC SIGNAL LINE	3  HAND CONTROL VALVE IN SIGNAL LINE

6.6 Symbols for self-actuated regulators, valves, and other devices (contd.)

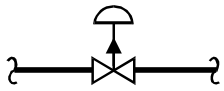
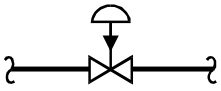
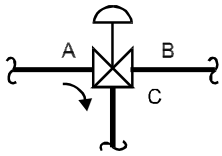
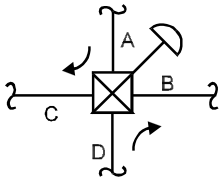
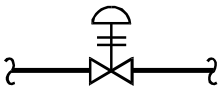
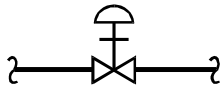
LEVEL	1  LEVEL REGULATOR WITH MECHANICAL LINKAGE	2	3
PRESSURE	1  PRESSURE-REDUCING REGULATOR, SELF-CONTAINED, WITH HANDWHEEL ADJUSTABLE SET POINT	2  PRESSURE-REDUCING REGULATOR WITH EXTERNAL PRESSURE TAP	3  DIFFERENTIAL-PRESSURE-REDUCING REGULATOR WITH INTERNAL AND EXTERNAL PRESSURE TAPS
	4  BACKPRESSURE REGULATOR, SELF-CONTAINED	5  BACKPRESSURE REGULATOR WITH EXTERNAL PRESSURE TAP	6  PRESSURE-REDUCING REGULATOR WITH INTEGRAL OUTLET PRESSURE RELIEF VALVE, AND OPTIONAL PRESSURE INDICATOR (TYPICAL AIR SET)
	7  PRESSURE RELIEF OR SAFETY VALVE, GENERAL SYMBOL	8  PRESSURE RELIEF OR SAFETY VALVE, STRAIGHT-THROUGH PATTERN, SPRING- OR WEIGHT-LOADED, OR WITH INTEGRAL PILOT	9  VACUUM RELIEF VALVE, GENERAL SYMBOL

6.6 Symbols for self-actuated regulators, valves, and other devices (contd.)



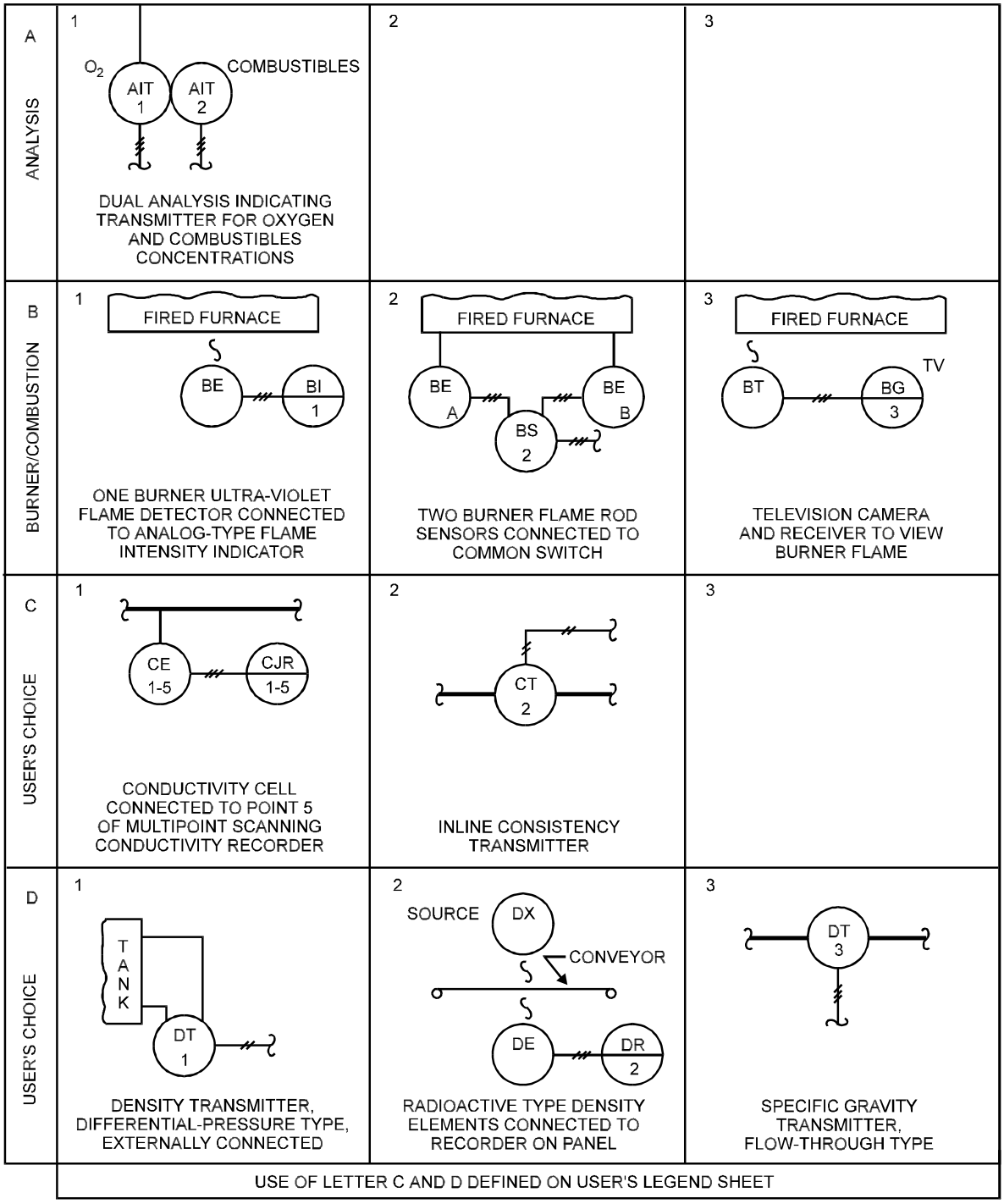
* The soleniod-tripped pressure relief valve is one of the class of power-actuated relief valves and is grouped with the other types of relief valves even though it is not entirely a self-actuated device.

6.7 Symbols for actuator action in event of actuator power failure (shown typically for diaphragm-actuated control valve).

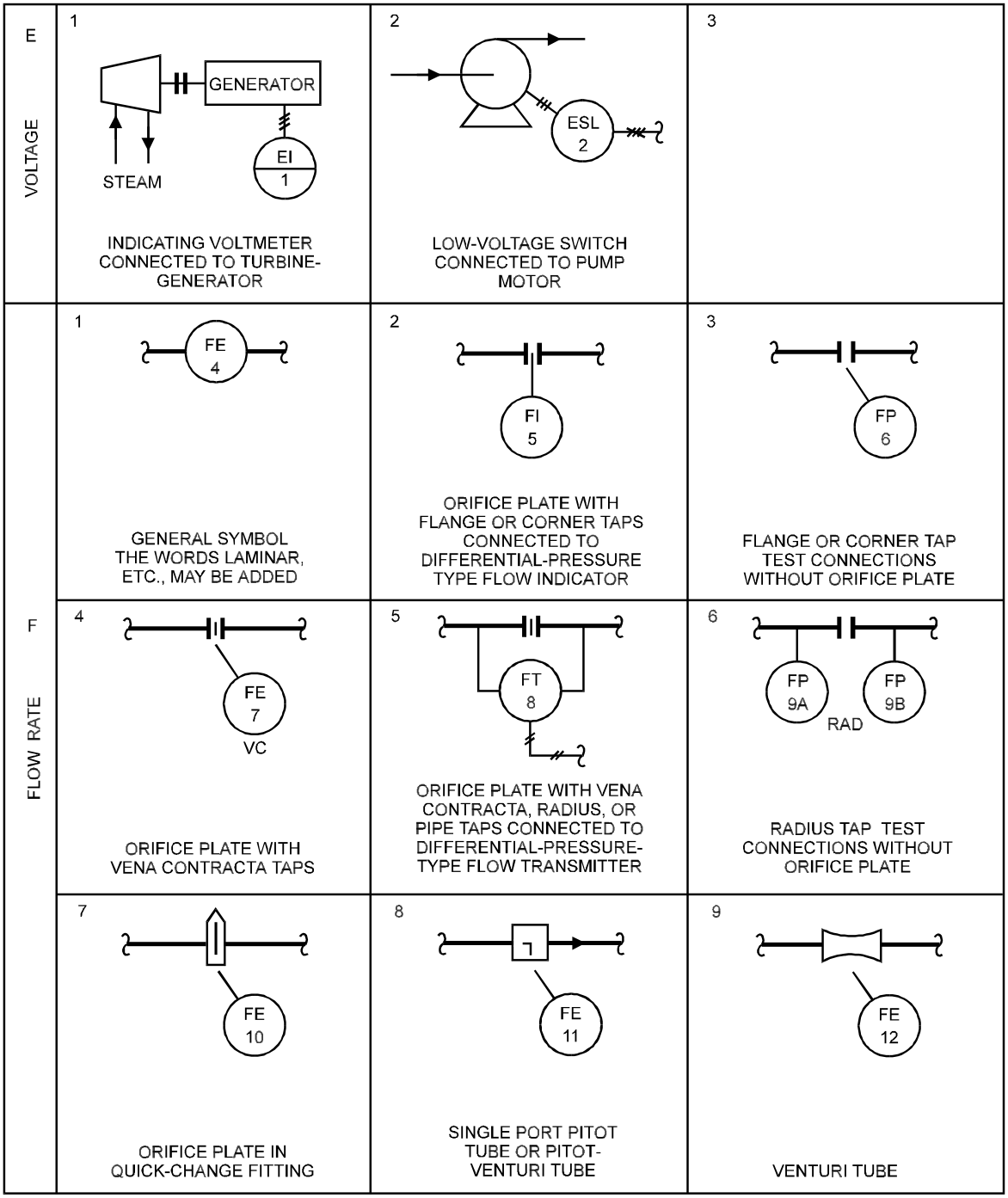
1  TWO-WAY VALVE, FAIL OPEN	2  TWO-WAY VALVE, FAIL CLOSED	3  THREE-WAY VALVE, FAIL OPEN TO PATH A-C
4  FOUR-WAY VALVE, FAIL OPEN TO PATHS A-C AND D-B	5  ANY VALVE, FAIL LOCKED (POSITION DOES NOT CHANGE)	6  ANY VALVE, FAIL INDETERMINATE

The failure modes indicated are those commonly defined by the term, "shelf-position". As an alternative to the arrows and bars, the following abbreviations may be employed:
FO - Fail Open
FC - Fail Closed
FL - Fail Locked (last position)
FI - Fail Indeterminate

6.8 Primary element symbols



6.8 Primary element symbols (contd.)

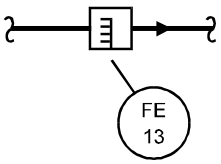


6.8 Primary element symbols (contd.)

F

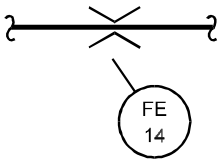
FLOW RATE (contd.)

10



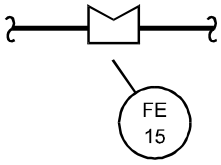
AVERAGING
PITOT TUBE

11



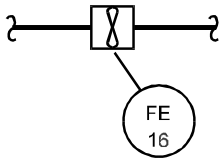
FLUME

12



WEIR

13



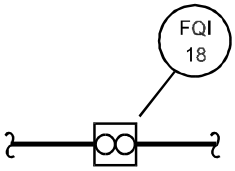
TURBINE-OR PROPELLER-
TYPE PRIMARY ELEMENT

14



VARIABLE AREA
FLOW INDICATOR

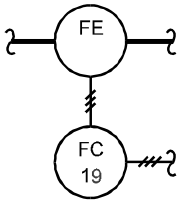
15



POSITIVE-DISPLACEMENT-
TYPE FLOW TOTALIZING
INDICATOR

16

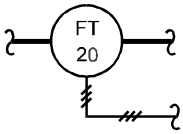
LAMINAR FLOW,
ETC.



FLOW ELEMENT WITH
CONNECTION FOR
CONTROLLER

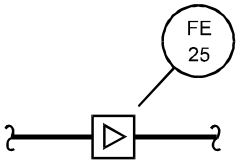
17

MASS FLOW
ETC.



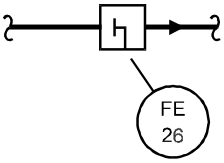
FLOW ELEMENT INTEGRAL
WITH TRANSMITTER

18



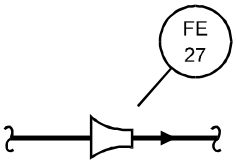
VORTEX SENSOR

19



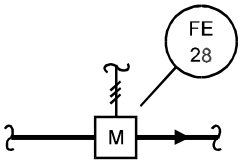
TARGET TYPE SENSOR

20



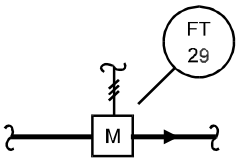
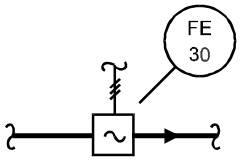
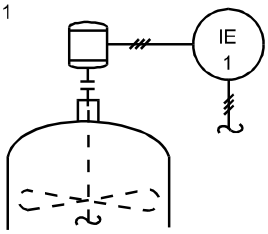
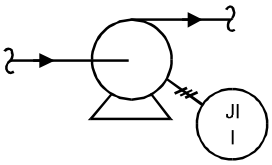
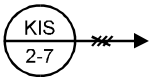
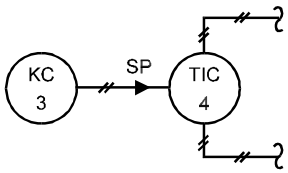
FLOW NOZZLE

21



MAGNETIC FLOWMETER

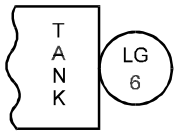
6.8 Primary element symbols (contd.)

F FLOW RATE (CONTD.)	22  MAGNETIC FLOWMETER WITH INTEGRAL TRANSMITTER	23  SONIC FLOWMETER "DOPPLER" OR "TRANSIT TIME" MAY BE ADDED	24
I CURRENT	1  CURRENT TRANSFORMER MEASURING CURRENT OF ELECTRIC MOTOR	2	3
J POWER	1  INDICATING WATTMETER CONNECTED TO PUMP MOTOR	2	3
K TIME OR TIME-SCHEDULE	1  CLOCK	2  MULTIPOINT ON-OFF TIME SEQUENCING PROGRAMMER POINT 7	3  TIME-SCHEDULE CONTROLLER, ANALOG TYPE, OR SELF-CONTAINED FUNCTION GENERATOR

6.8 Primary element symbols (contd.)

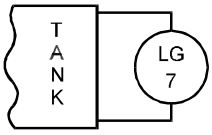
L LEVEL

1



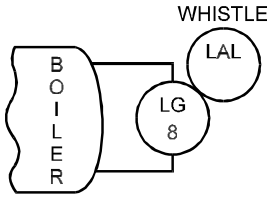
GAGE GLASS, INTEGRALLY MOUNTED ON TANK

2



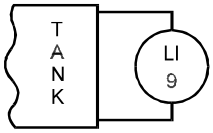
GAGE GLASS, EXTERNALLY CONNECTED

3



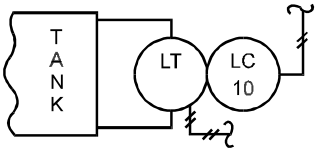
WATER COLUMN WITH INTEGRAL GAGE GLASS AND ALARM WHISTLE

4



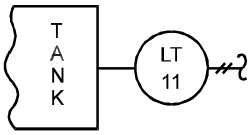
LEVEL INDICATOR, WITH TWO CONNECTIONS

5



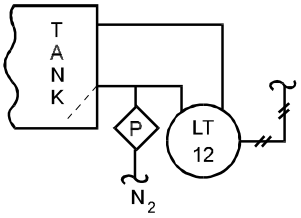
DUPLEX LEVEL TRANSMITTER-CONTROLLER, WITH TWO CONNECTIONS

6



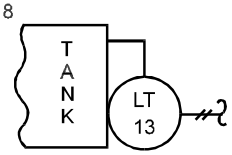
LEVEL TRANSMITTER, WITH ONE CONNECTION

7



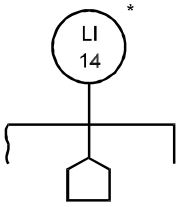
LEVEL TRANSMITTER, DIFFERENTIAL-PRESSURE TYPE, EXTERNALLY CONNECTED, WITH DIP TUBE

8



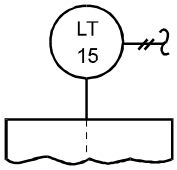
LEVEL TRANSMITTER, DIFFERENTIAL-PRESSURE TYPE, MOUNTED ON TANK

9



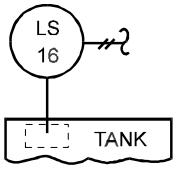
LEVEL INDICATOR, FLOAT TYPE

10



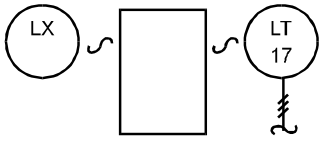
CAPACITANCE OR DIELECTRIC TYPE LEVEL ELEMENT CONNECTED TO LEVEL TRANSMITTER (TAG LEVEL ELEMENT LE-15)

11



LEVEL SWITCH, PADDLE WHEEL OR LEVER TYPE, TO MEASURE LEVEL OF SOLIDS

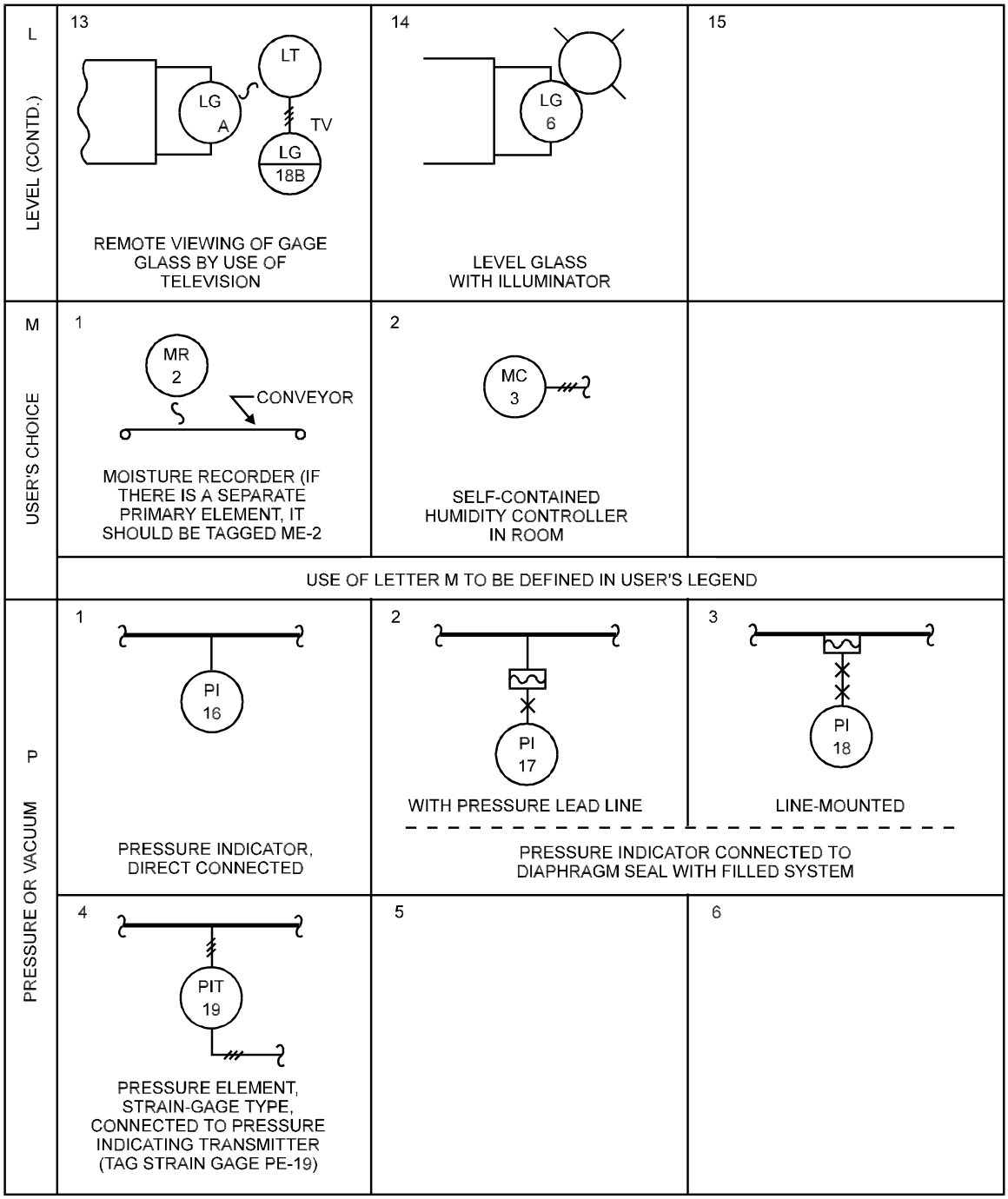
12



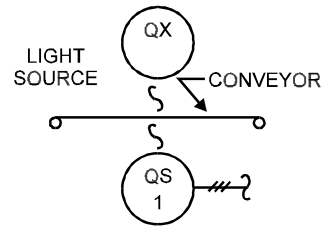
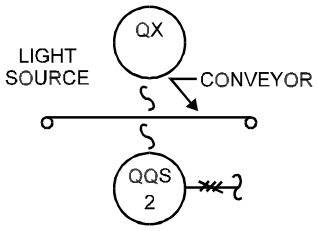
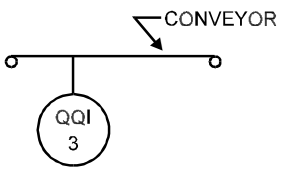
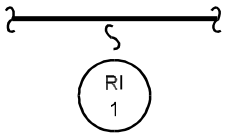
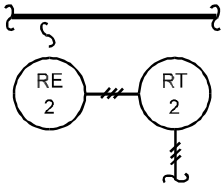
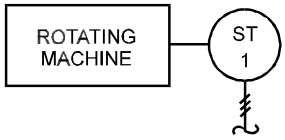
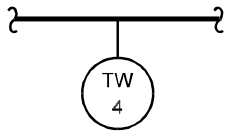
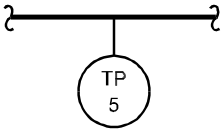
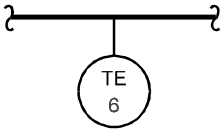
RADIOACTIVE- OR SONIC-TYPE LEVEL TRANSMITTER WITH INTEGRAL SENSOR

* Notation such as "mounted at grade" may be added.

6.8 Primary element symbols (contd.)



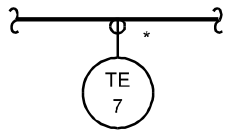
6.8 Primary element symbols (contd.)

Q QUANTITY	1  COUNTING SWITCH, PHOTO-ELECTRIC TYPE, WITH SWITCH ACTION FOR EACH EVENT	2  COUNTING SWITCH, PHOTO-ELECTRIC TYPE, WITH SWITCH ACTION BASED ON CUMULATIVE TOTAL	3  INDICATING COUNTER, MECHANICAL TYPE
R RADIATION	1  RADIATION INDICATOR	2  RADIATION MEASURING ELEMENT AND TRANSMITTER	3
S SPEED OR FREQUENCY	1  SPEED TRANSMITTER	2	3
T TEMPERATURE	1  TEMPERATURE CONNECTION WITH WELL	2  TEMPERATURE TEST CONNECTION WITHOUT WELL	3  TEMPERATURE ELEMENT WITHOUT WELL (ELEMENT NOT CONNECTED TO SECONDARY INSTRUMENT)

6.8 Primary element symbols (contd.)

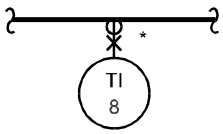
T TEMPERATURE (CONTD.)

4



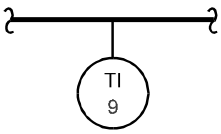
TEMPERATURE ELEMENT WITH WELL (ELEMENT NOT CONNECTED TO SECONDARY INSTRUMENT)

5



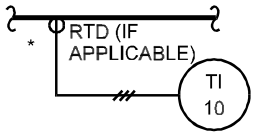
FILLED-SYSTEM-TYPE TEMPERATURE INDICATOR WITH WELL

6



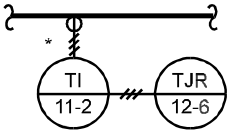
BIMETALLIC-TYPE THERMOMETER, GLASS THERMOMETER, OR OTHER LOCAL UNCLASSIFIED TEMPERATURE INDICATOR

7



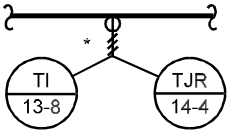
THERMOCOUPLE, RESISTANCE BULB (RTD) OR THERMISTOR (TH) CONNECTED TO TEMPERATURE INDICATOR (TAG ELEMENT TE-10)

8



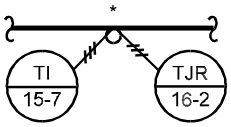
THERMOCOUPLE CONNECTED TO MULTIPOINT INDICATOR RECORDING ON MULTIPOINT SCANNING RECORDER (TAG ELEMENT TE-11-2)

9



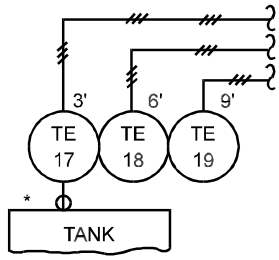
THERMOCOUPLE PARALLEL-WIRED TO MULTIPOINT INDICATOR AND MULTIPOINT SCANNING RECORDER (TAG ELEMENT TE-13-8/14-4)

10



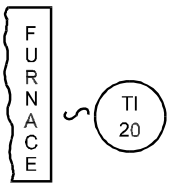
DUAL OR DUPLEX THERMOCOUPLE CONNECTED TO MULTIPOINT INDICATOR AND MULTIPOINT SCANNING RECORDER (TAG ELEMENT TE-15-7/16-2)

11



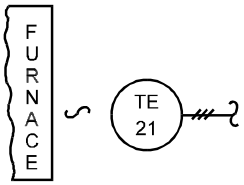
MULTI-ELEMENT THERMOCOUPLE FOR DIFFERENT ELEVATIONS, WITH WELL IN TANK

12



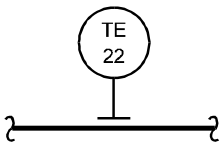
THERMAL-RADIATION TYPE TEMPERATURE INDICATOR, SELF-CONTAINED

13



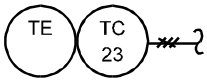
THERMAL-RADIATION-TYPE TEMPERATURE ELEMENT

14



SURFACE-MOUNTED TEMPERATURE SENSOR

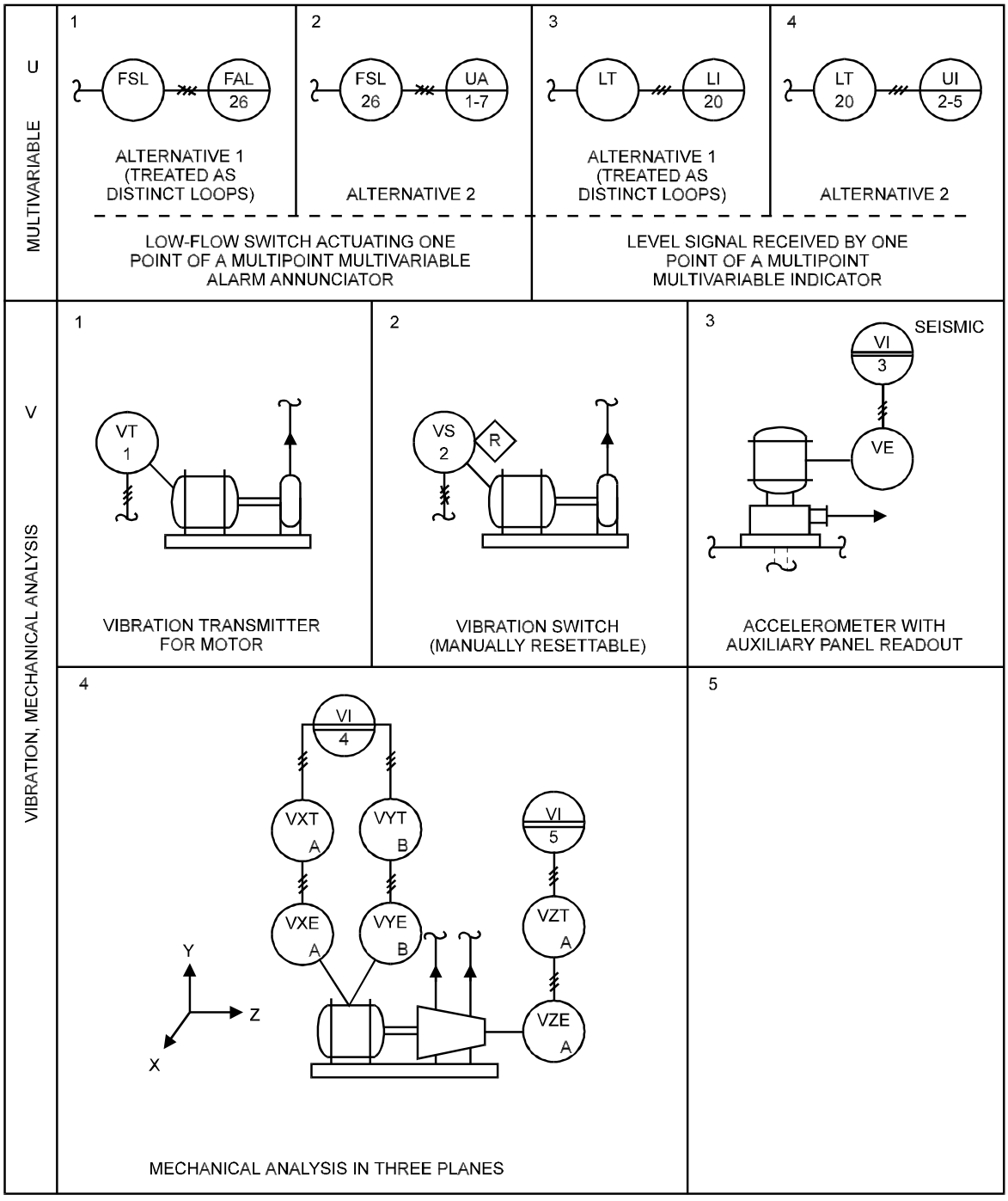
15



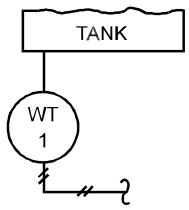
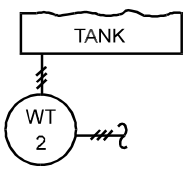
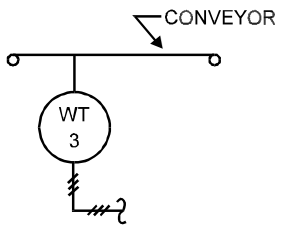
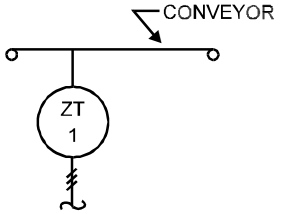
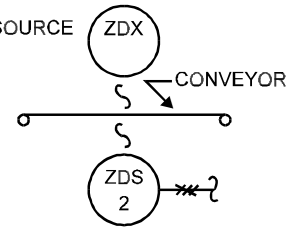
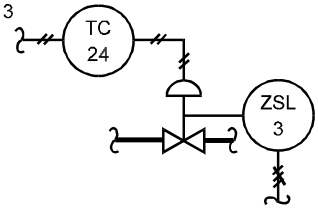
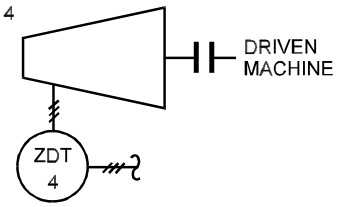
THERMOSTAT SENSING AMBIENT TEMPERATURE

* Use of the thermowell symbol is optional. However, ues or omission of the symbol should be consistent through a project.

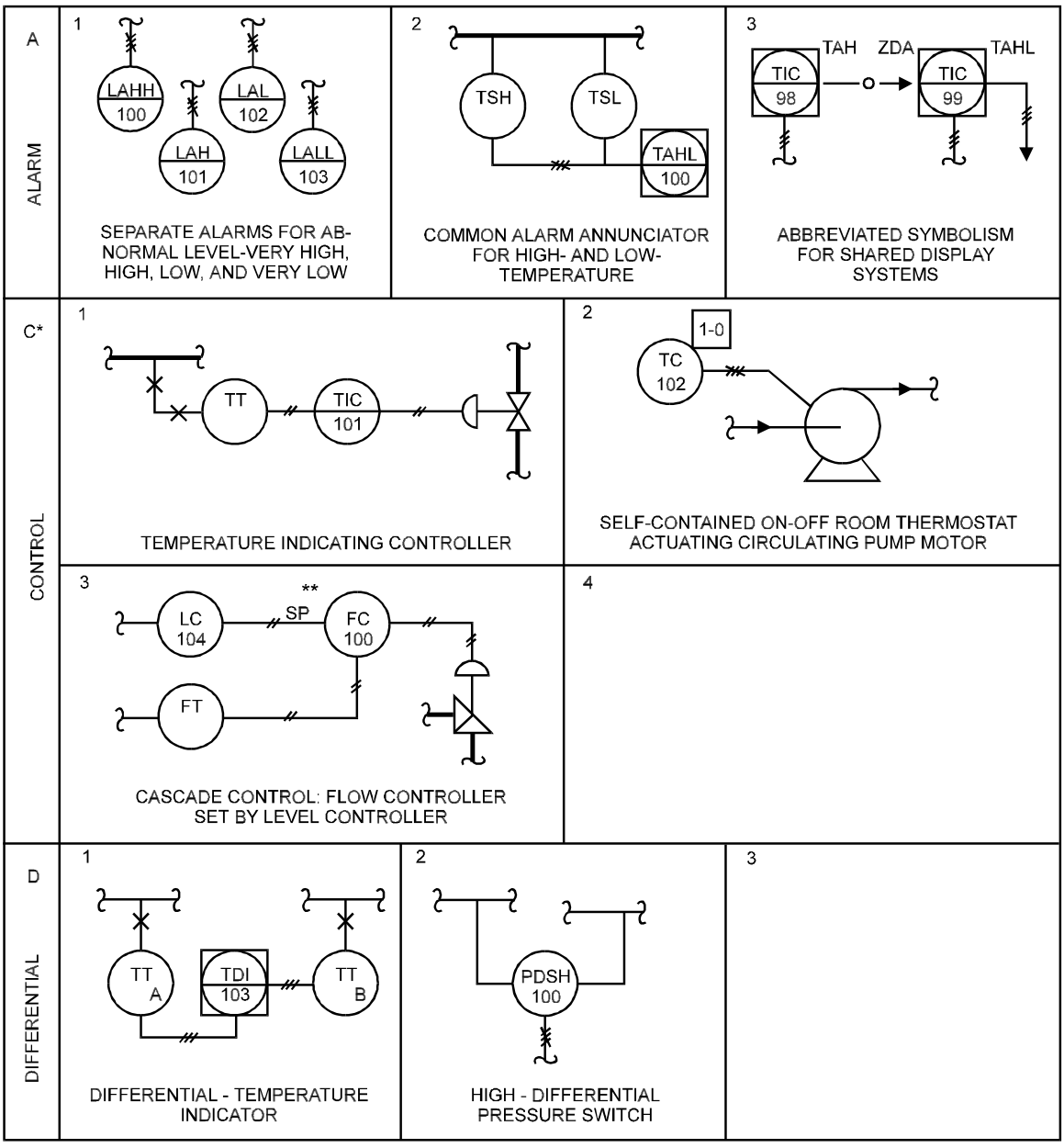
6.8 Primary element symbols (contd.)



6.8 Primary element symbols (contd.)

W WEIGHT OR FORCE	1  WEIGHT TRANSMITTER, DIRECT-CONNECTED	2  STRAIN GAGE CONNECTER TO SEPARATE WEIGHT TRANSMITTER (TAG STRAIN GAGE WE-2)	3  WEIGH-BELT SCALE TRANSMITTER
Z POSITION, DIMENSION	1  ROLL-THICKNESS TRANSMITTER	2  THICKNESS SWITCH, RADIOACTIVE TYPE	3  LIMIT SWITCH THAT IS ACTUATED WHEN VALVE IS CLOSED TO A PRE-DETERMINED POSITION
	4  TURBINE SHELL/ROTOR DIFFERENTIAL EXPANSION TRANSMITTER (TAG PRIMARY ELEMENT ZDE-4)	5	6

6.9 Examples — functions



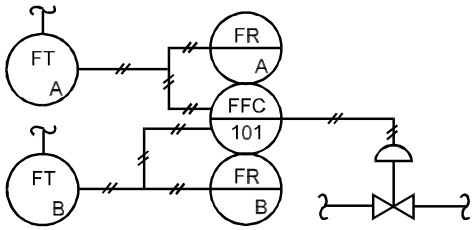
* It is expected that control modes will not be designed on a diagram. However, designations may be used outside the controller symbol, if desired, in combinations such as [%], [∫], [1-0].

** A controller is understood to have integral manual set-point adjustment unless means of remote adjustment is indicated. The remote set-point designation is SP.

6.9 Examples — functions (contd.)

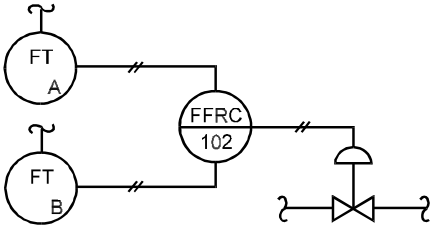
F RATIO

1



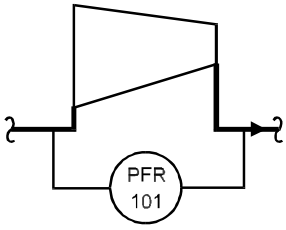
FLOW-RATIO CONTROLLER WITH TWO PENS TO RECORD FLOW

2



FLOW-RATIO CONTROLLER WITH ONE PEN TO RECORD FLOW-RATIO

3

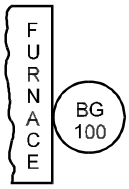


DIRECT-CONNECTED COMPRESSION-RATIO RECORDER

4

G VIEWING DEVICE, GLASS

1



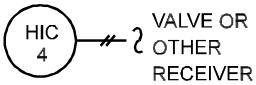
SIGHT GLASS FOR INTERNAL VIEWING

2

3

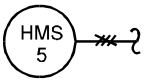
H HAND

1



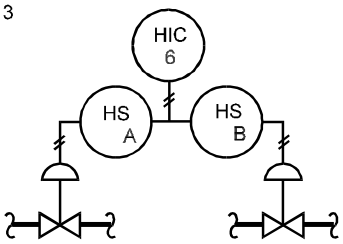
MANUAL LOADING STATION WITH OUTPUT GAGE

2



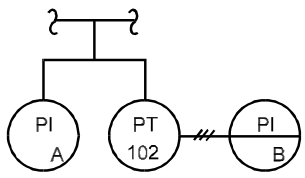
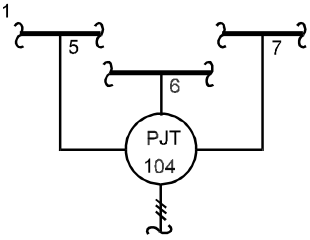
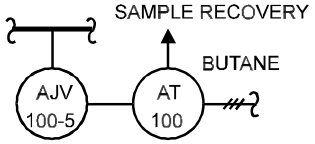
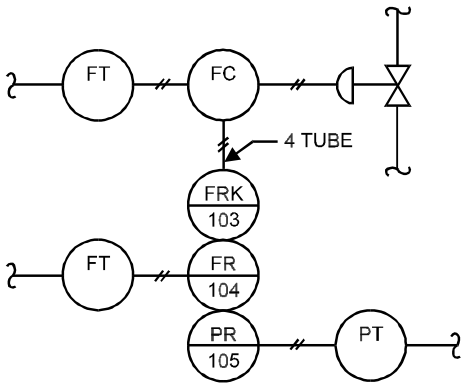
HAND-ACTUATED ELECTRIC SWITCH, MOMENTARY

3

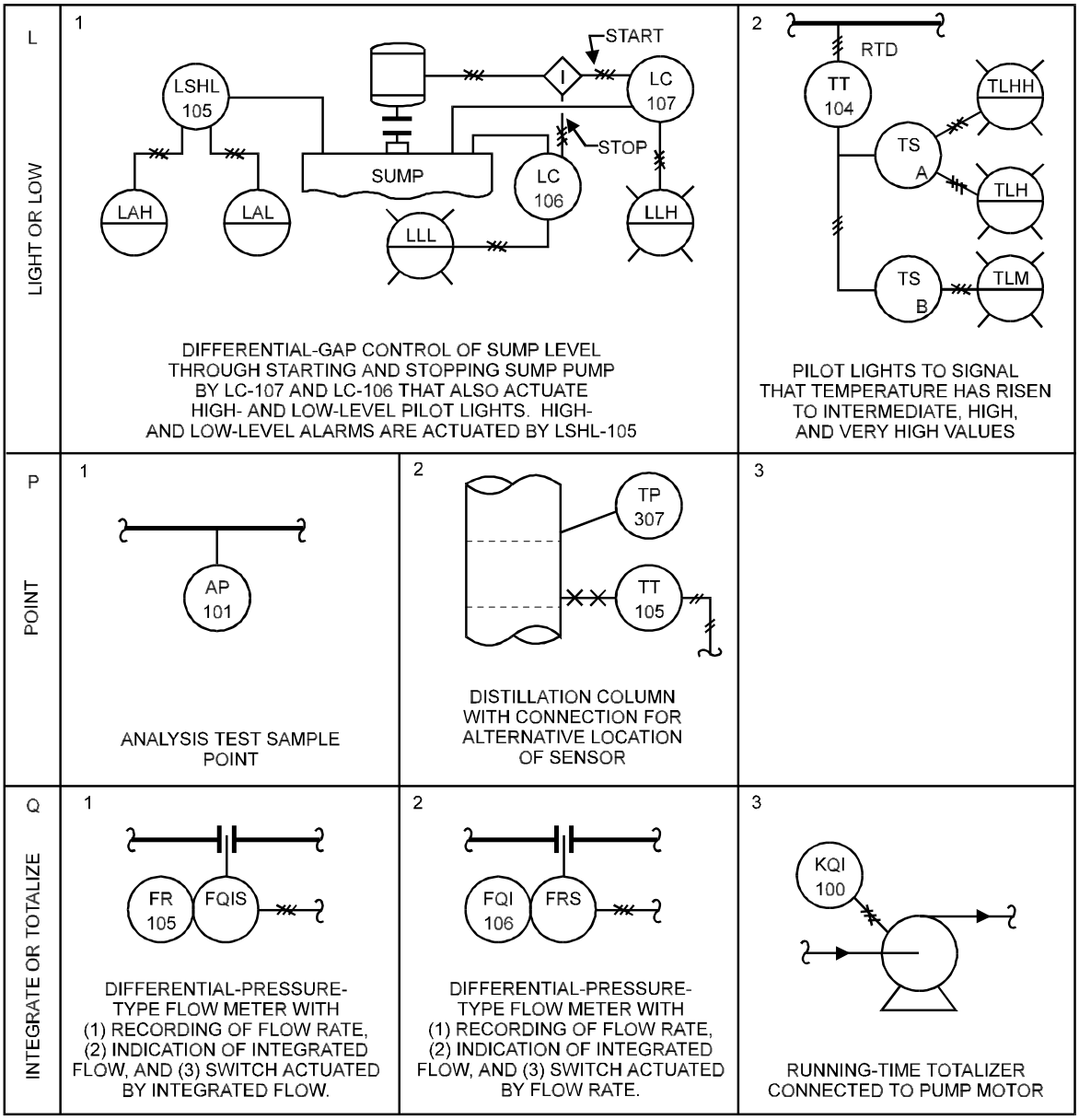


MANUAL LOADING STATION WITH HAND ACTUATED SWITCHES

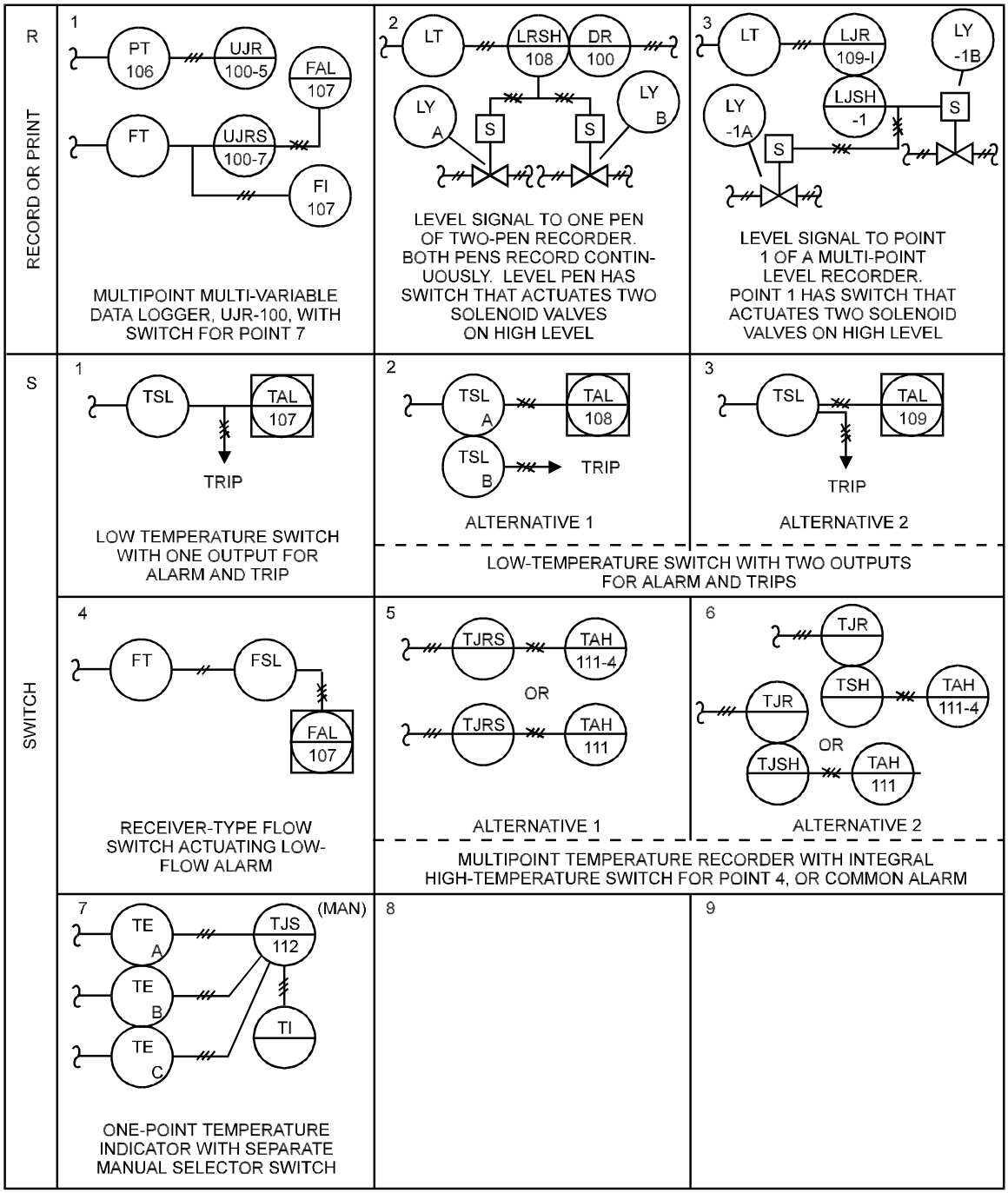
6.9 Examples — functions (contd.)

I INDICATE	1  LOCAL PRESSURE INDICATOR AND PRESSURE TRANSMITTER WITH COMMON TAP AND PANEL-MOUNTED PRESSURE INDICATOR	2	3
J SCAN	1  PRESSURE-SCANNING TRANSMITTER CONNECTED TO PROCESS POINT 5,6,7	2  SAMPLE LINE CONNECTED TO BUTANE-CONCENTRATION TRANSMITTER THROUGH EXTERNAL SAMPLE-SCANNING VALVE	3
K CONTROL STATION	1  RECORDING FLOW CONTROL STATION PANEL MOUNTED, WITH ADDITIONAL FLOW AND PRESSURE PENS, AND LOCAL CONTROLLER	2	

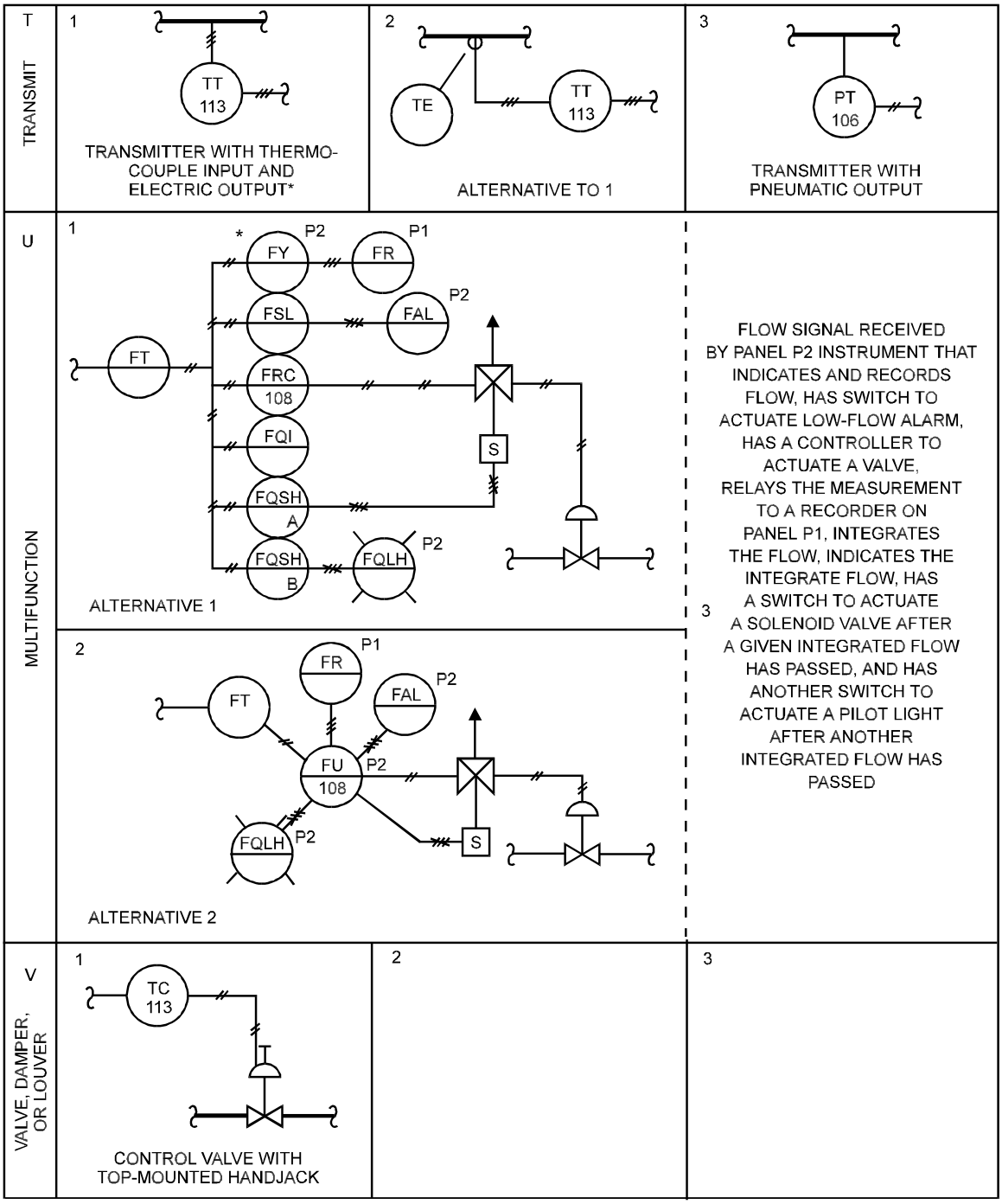
6.9 Examples — functions (contd.)



6.9 Examples — functions (contd.)

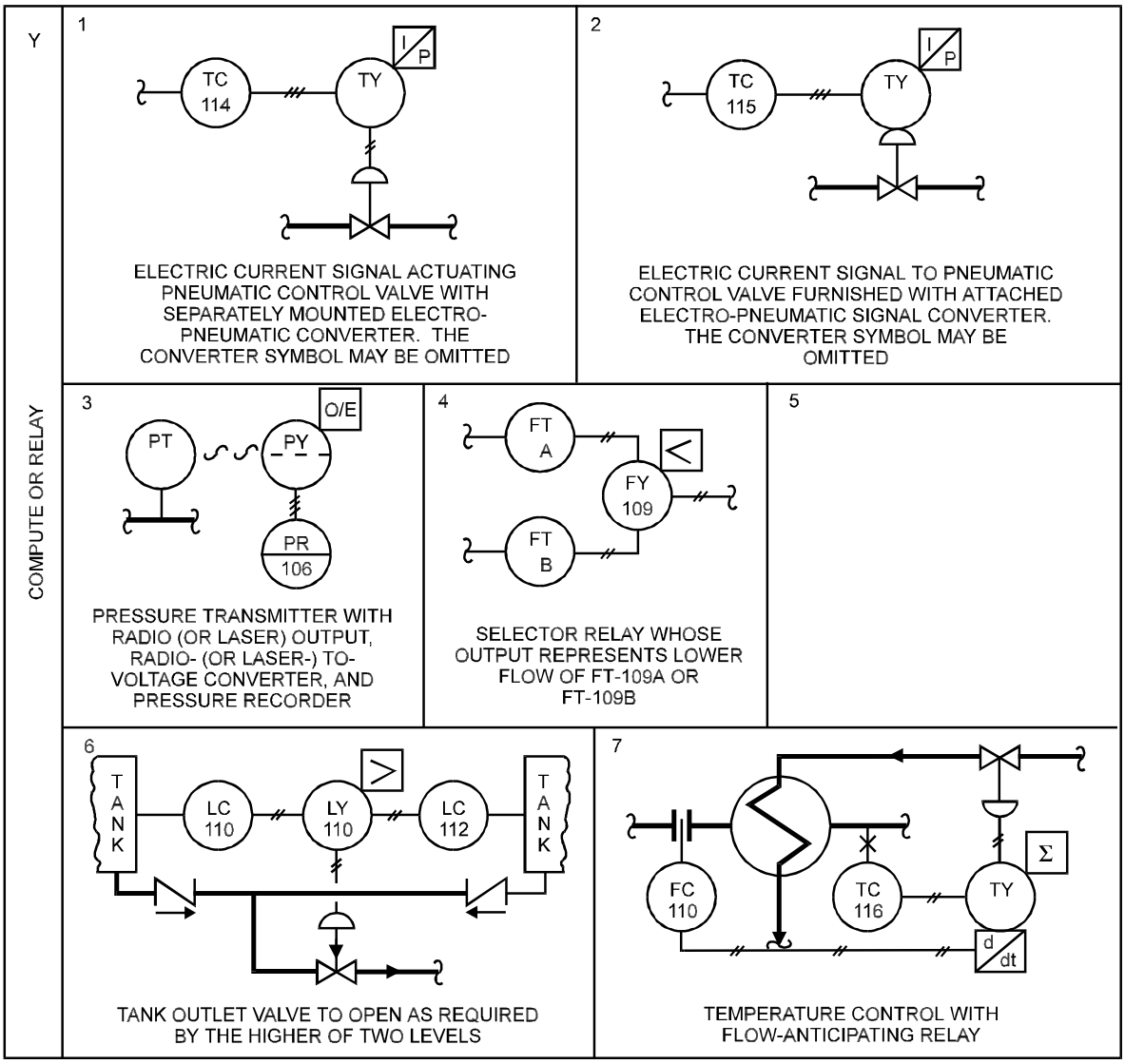


6.9 Examples — functions (contd.)

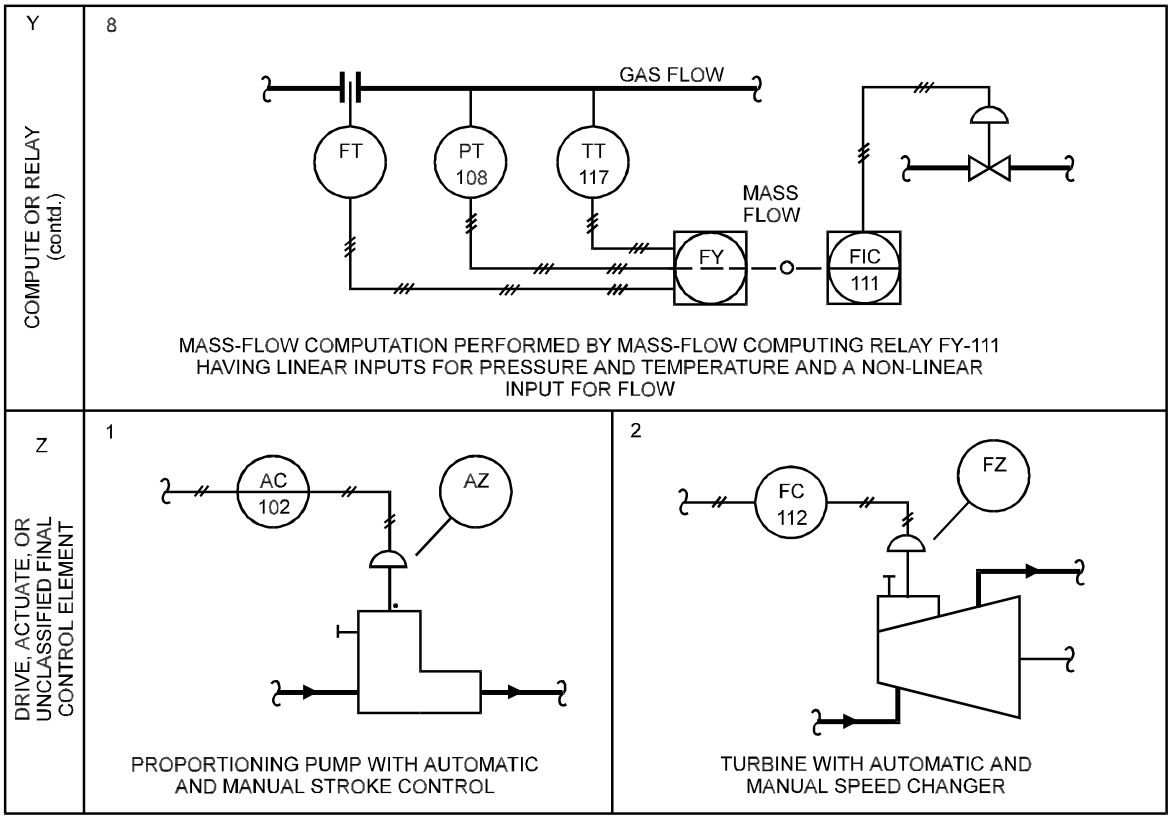


* See definition of converter verses transmitter.

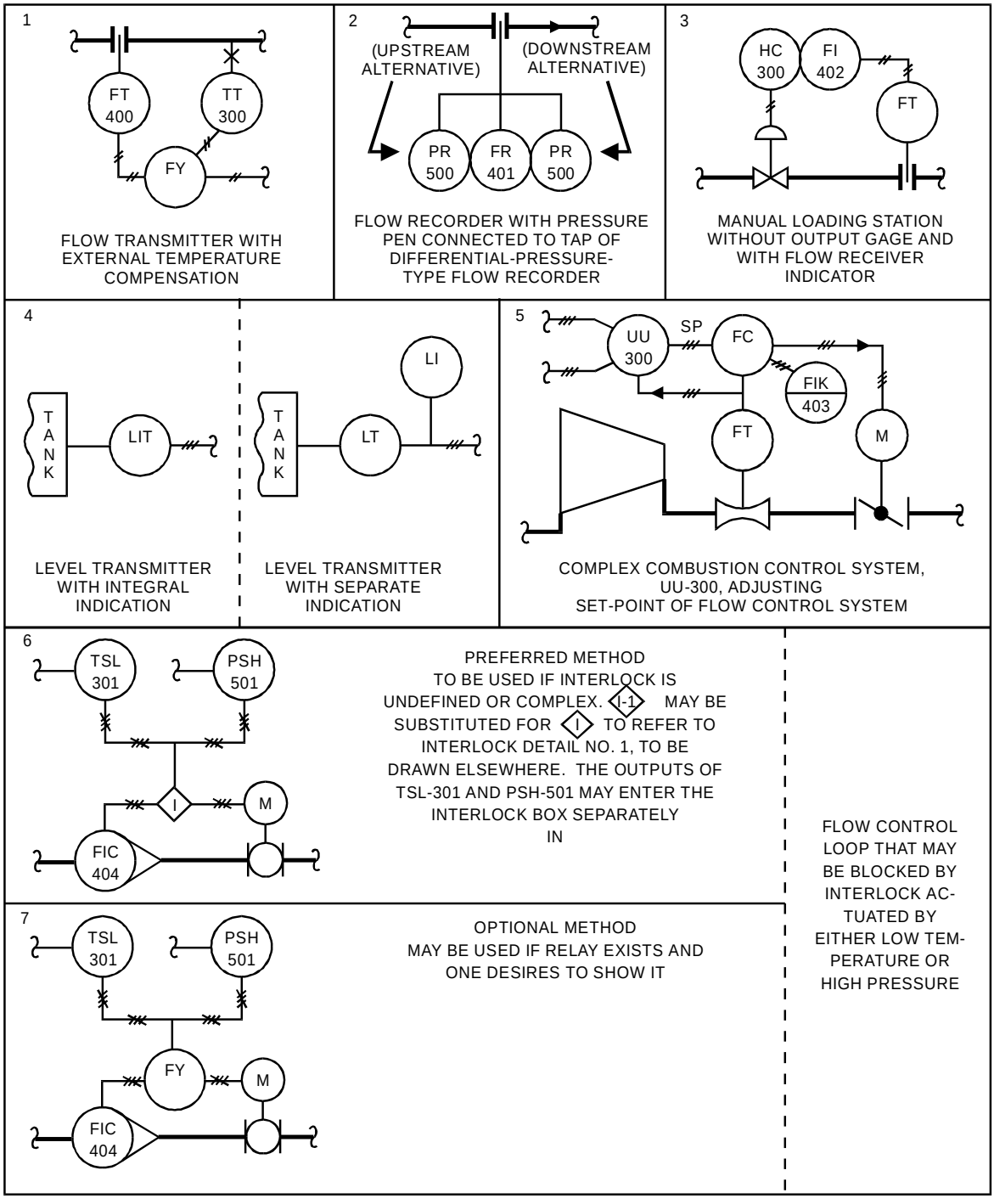
6.9 Examples — functions (contd.)



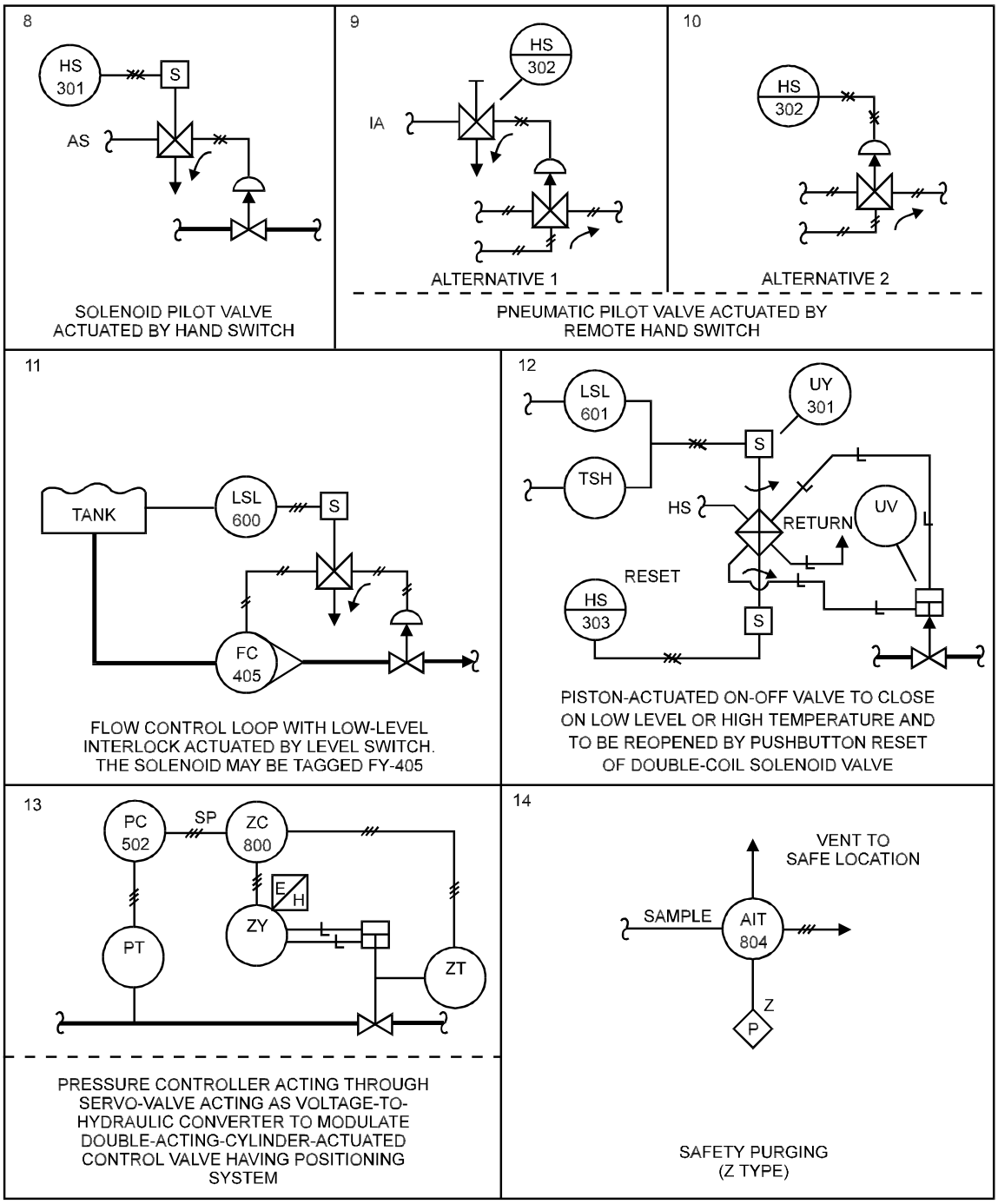
6.9 Examples — functions (contd.)



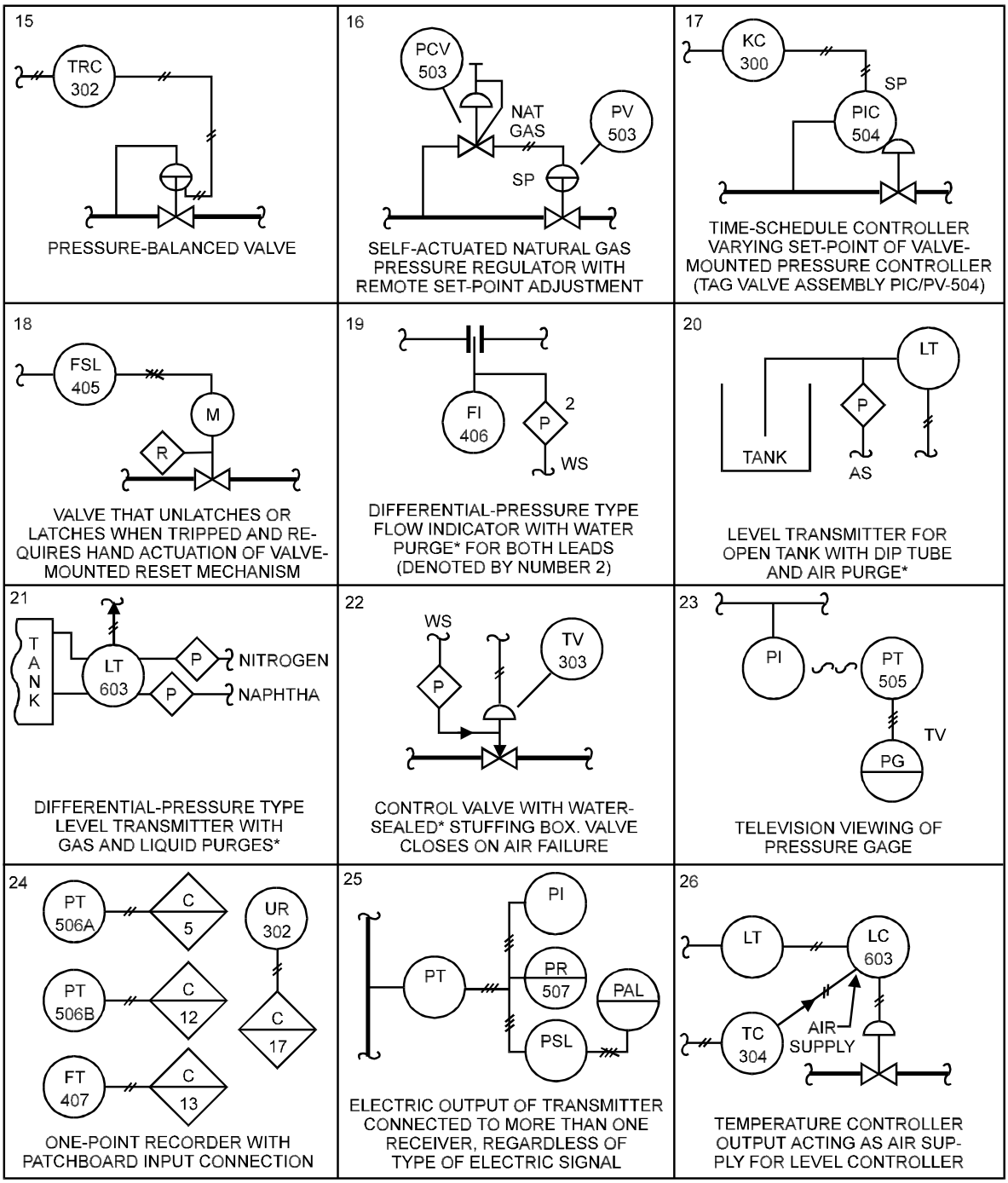
6.10 Examples — miscellaneous combinations



6.10 Examples — miscellaneous combinations (contd.)



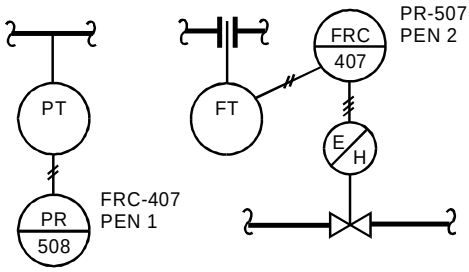
6.10 Examples — miscellaneous combinations (contd.)



* The purge fluid supplies may use the same abbreviations as the instrument power supplies.

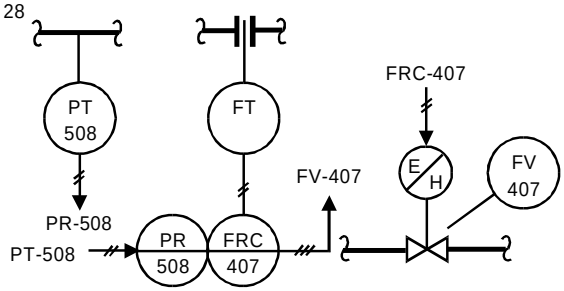
6.10 Examples — miscellaneous combinations (contd.)

27



ALTERNATIVE 1

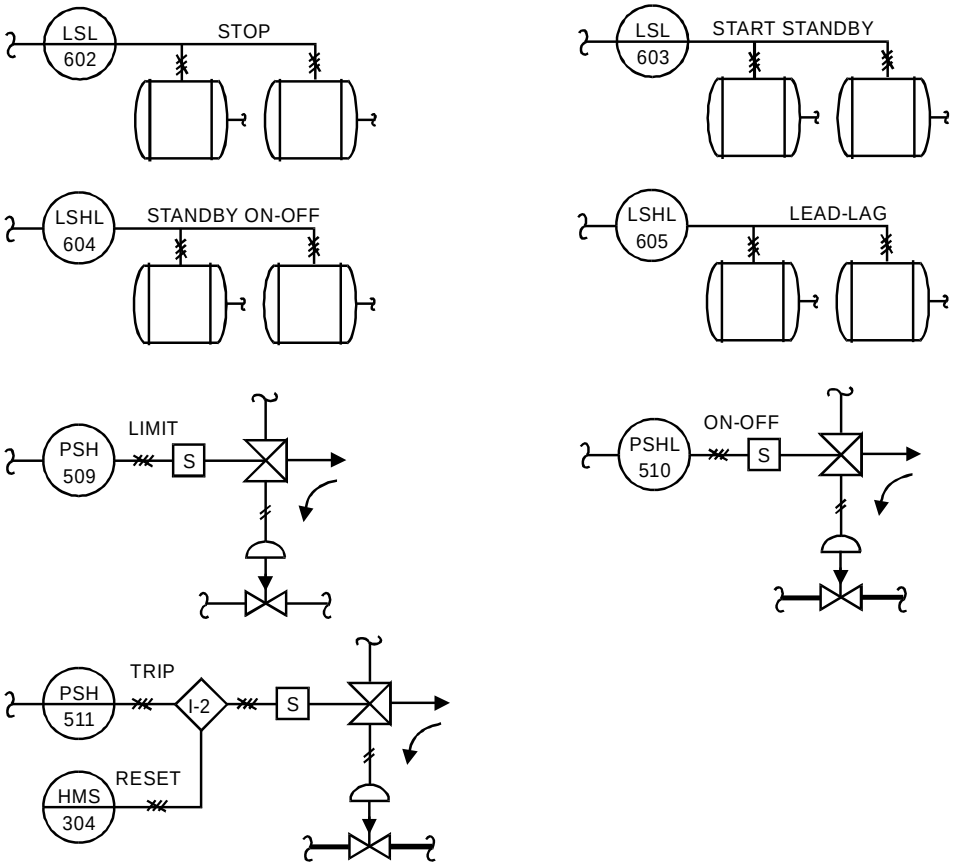
28



ALTERNATIVE 2

INTERRELATED INSTRUMENTS WHOSE SYMBOLS ARE SCATTERED ON DIAGRAM. (A MULTI-POINT INSTRUMENT, SUCH AS A DATA LOGGER, THAT IS DESIGNATED WITH POINT NUMBERS ON A DIAGRAM IS NOT EXPECTED TO HAVE THE SYMBOLS FOR THE VARIOUS POINTS TIED OR REFERENCED TOGETHER). PEN ASSIGNMENTS NEED NOT BE SHOWN ON A DIAGRAM IF IT IS THE USER'S PREFERENCE TO SHOW THEM IN AN INDEX.

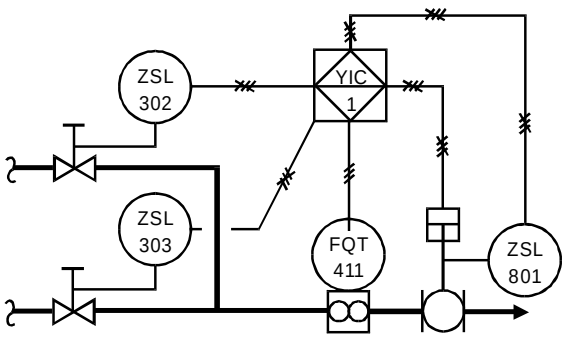
29



THE JUDICIOUS USE OF WORDS CAN CLARIFY DESIGN INTENT

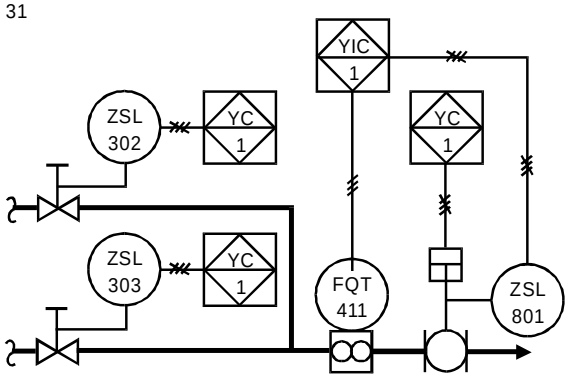
6.10 Examples — miscellaneous combinations (contd.)

30



ALTERNATIVE 1

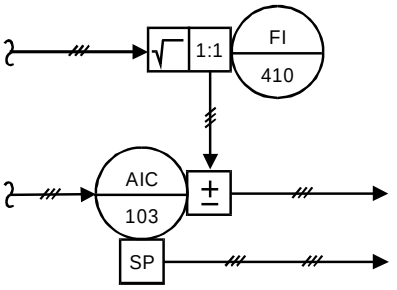
31



ALTERNATIVE 2

PROGRAMMABLE CONTROLLER SCHEME. ALTERNATIVE 2 ALLOWS MORE DEFINITION OF WHICH SIGNALS ARE ACCESSIBLE TO THE OPERATOR

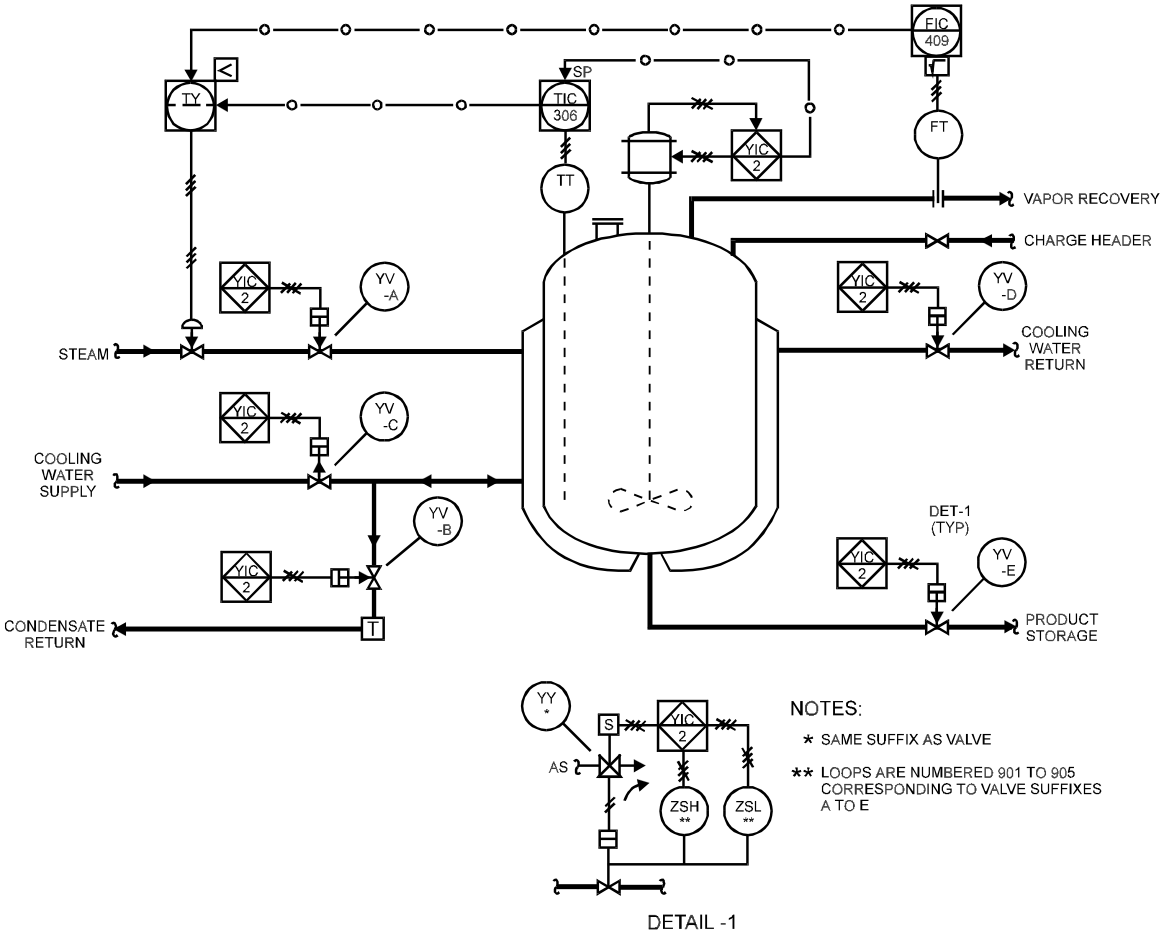
32



INDIVIDUAL INSTRUMENTS WITH INTEGRAL FUNCTION BLOCKS: SQUARE ROOT EXTRACTION, 1:1 FEED BOOST, FEED FORWARD BIAS ON CONTROLLER OUTPUT, SETPOINT OUTPUT SIGNAL

33

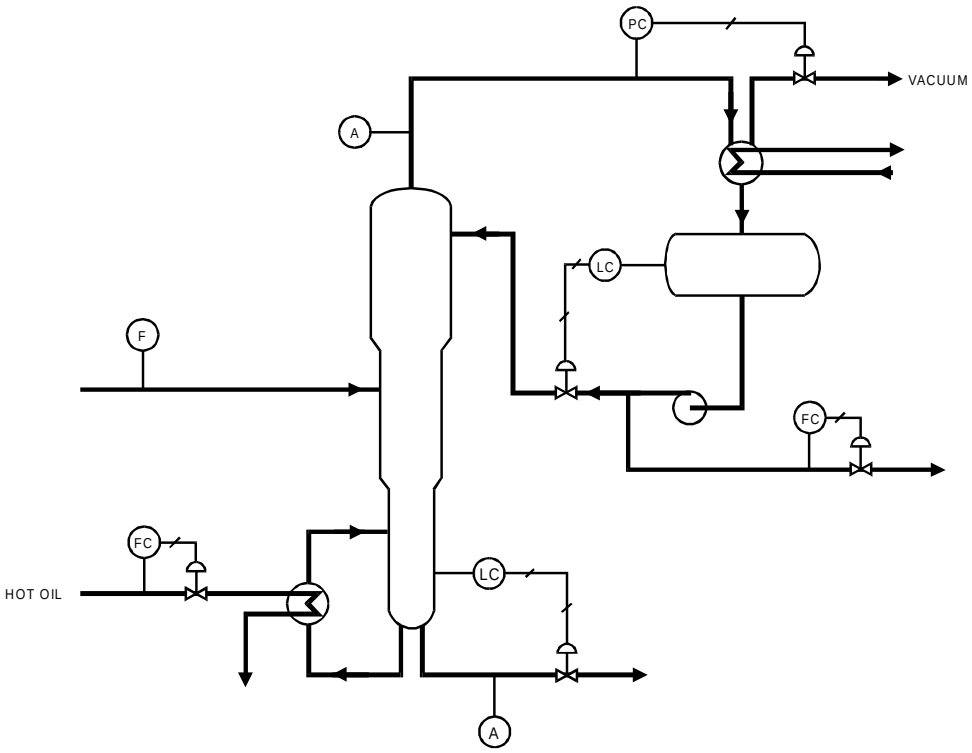
6.11 Example — complex combinations



Example of complex batch kettle control involving both shared display/ control and programmable logic control. The purpose of this drawing is to allow a general understanding of the control scheme and to define and identify the connection components. Detailed understanding would be obtained from the study of other documents.

6.12 Example — degree of detail

1 TYPICAL SYMBOLISM FOR SIMPLIFIED DIAGRAMS

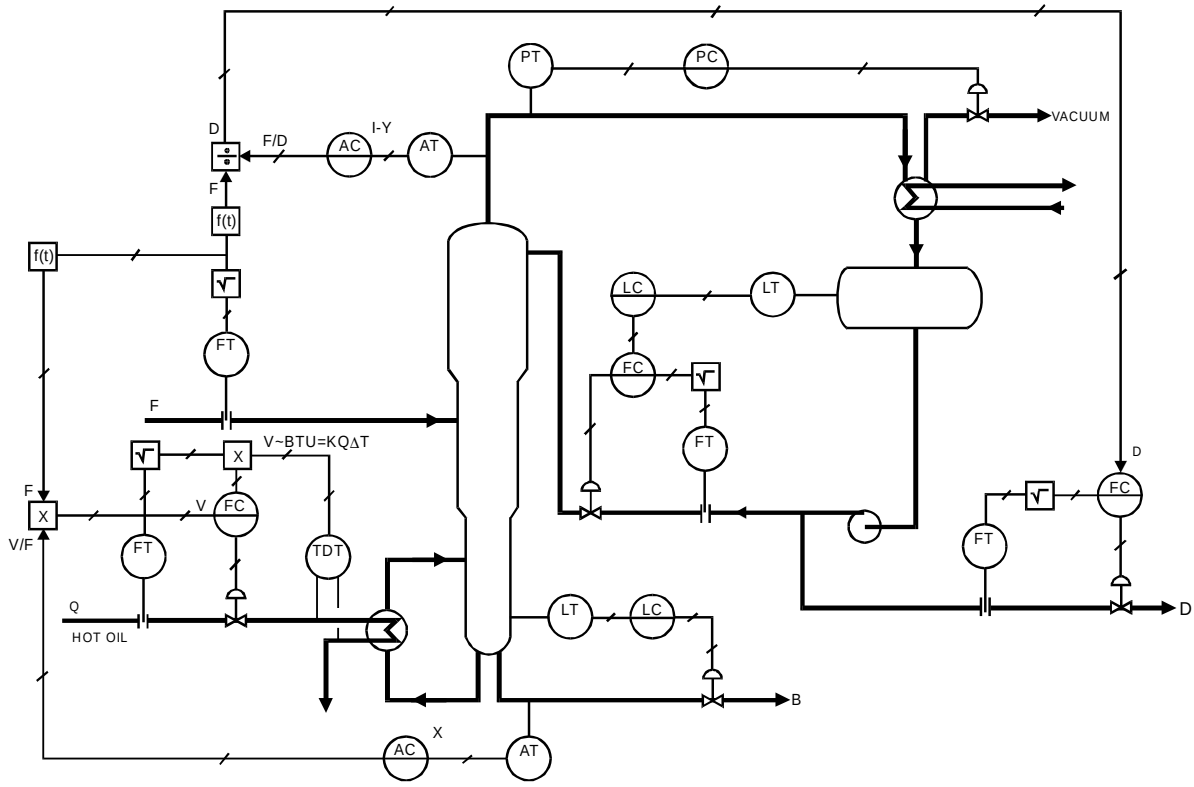


Simplified symbolism and abbreviated identification used to define the principal points of measurement and control interest.

* SEE SECTION 4.4 FOR DISCUSSION

6.12 Example — degree of detail (contd.)

2 TYPICAL SYMBOLISM FOR CONCEPTUAL DIAGRAMS

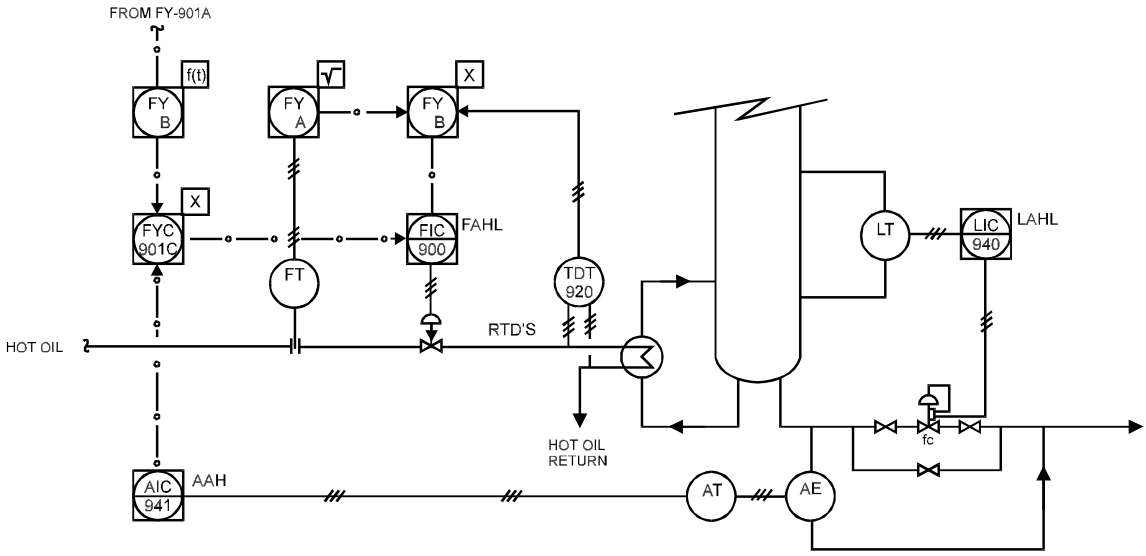


Functionally oriented symbolism and abbreviated identification used to develop control concepts without concern for specific hardware.

* SEE SECTION 4.4 FOR DISCUSSION

6.12 Example — degree of detail (contd.)

3 TYPICAL SYMBOLISM FOR DETAILED DIAGRAMS



Detailed symbolism and more complete identification used to describe the control system when type of hardware and kinds of signals have been chosen.

* SEE SECTION 4.4 FOR DISCUSSION

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